

Language Models for Dementia Detection

Sprachmodelle für Demenzerkennung

Master's Thesis in in Medical Engineering

submitted
by

Julian Fritsch

born April 16, 1991 in Roth

Written at

Lehrstuhl für Mustererkennung (Informatik 5)

Department Informatik

Friedrich-Alexander-Universität Erlangen-Nürnberg

Advisors: Prof. Dr.-Ing. Elmar Nöth, PD Dr.-Ing. Tino Haderlein

Started: August 15, 2017

Finished: February 15, 2018

Ich versichere, dass ich die Arbeit ohne fremde Hilfe und ohne Benutzung anderer als der angegebenen Quellen angefertigt habe und dass die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegen hat und von dieser als Teil einer Prüfungsleistung angenommen wurde. Alle Ausführungen, die wörtlich oder sinngemäß übernommen wurden, sind als solche gekennzeichnet.

Die Richtlinien des Lehrstuhls für Studien- und Diplomarbeiten habe ich gelesen und anerkannt, insbesondere die Regelung des Nutzungsrechts.

Erlangen, den 15. Februar 2018

Übersicht

Die heutige Gesellschaft altert. Somit steigt auch die Zahl der Fälle neurodegenerativer Krankheiten, wie Demenz. Da bis heute kein Heilmittel existiert, sind verlässliche Tools zur automatischen Erkennung und Überwachung von Demenz notwendig. Sprache ist hierfür ein sowohl nützlicher als auch sinnvoller Indikator. Der kognitive Zustand einer Person kann anhand von Aufgaben erfasst werden, die auf spontaner Sprache basieren, z.B. die Beschreibung des Cookie-Theft-Bildes. Allerdings veranlasst Demenz Menschen dazu, Sprache in unvorhersehbarer Weise zu verwenden, was wiederum die Leistung eines Spracherkenners verschlechtert. Diese Arbeit stellt eine Methode vor, um die Qualität eines allgemeinen Spracherkenners für Deutsch zu verbessern, indem sein Sprachmodell an einen speziellen Anwendungsfall angepasst wird. Dafür werden ursprünglich englische und maschinell ins Deutsche übersetzte Transkriptionen von Cookie-Theft-Bildbeschreibungen des Pitt Corpus der DementiaBank Datenbank verwendet. Diese Anpassung wird anhand von 45 Aufnahmen deutscher Cookie-Theft-Beschreibungen evaluiert. Des weiteren wird gezeigt, dass Sprachmodelle, erstellt mit dem Toolkit *rwthlm*, zur Sprachmodellierung mit Neuronalen Netzen und Long-Short-Term-Memory-Zellen auch dazu verwendet werden können zu erkennen, ob ein Sprecher an Demenz leidet oder nicht: Mit einer Auswahl von insgesamt 499 Cookie Theft Bildbeschreibungen des Pitt Corpus werden jeweils Sprachmodelle der Alzheimer- und Kontrollgruppe erstellt. Jede Transkription wird auf beiden Modellen evaluiert, indem deren Perplexität berechnet wird. Dies geschieht in einem Kreuzvalidierungsverfahren. Allein diese zwei Perplexitätswerte weisen genug Diskrepanz auf, um damit Alzheimerpatienten von der Kontrollgruppe mit einer Genauigkeit von 85.6% bei Gleichfehlerrate zu unterscheiden.

Abstract

In today's aging society, the number of neurodegenerative diseases, such as dementia, increases. Since currently no cure exists, reliable tools for automatic diagnosis and monitoring of dementia are necessary. For that, language is an indicator which is useful and reasonable. A person's cognitive state can be assessed by tasks that are based on spontaneous speech, e. g. describing the Cookie Theft picture. However, dementia's pathology causes persons to use unforeseen language which degrades the performance of automatic speech recognition. This thesis first presents a way to improve speech recognition performance by adapting a German large-vocabulary continuous speech recognizer's language model to a specific domain by machine translated native English transliterations of that domain from DementiaBank's Pitt corpus. This adaptation is evaluated on a dataset of 45 German recordings of Cookie Theft picture descriptions. Furthermore, the *rwthlm* toolkit is used to build deep neural network language models with long short term-memory cells. These language models are used to detect if a person suffers from dementia or not: a total of 499 transliterations selected from the Pitt corpus are used to build language models of all control and all Alzheimer speakers. Every transliteration is evaluated on those two models in terms of perplexity in a leave-one-speaker-out cross-validation scheme. The resulting perplexity values reveal enough discrepancy to classify patients on just those two values with an accuracy of 85.6% at equal error rate.

Acknowledgments

The experiments carried out in this thesis would not have been possible without the contributions of many people. I would like to thank those who supported me: Tanja Richter-Schmidinger from Department of Psychiatry and Psychotherapy, University Clinic Erlangen for providing recordings and associated anonymized annotation, Silvio Patruno, Andreas Ratzka and Wei Zhou from EML European Media Laboratory GmbH for allowing us to use their services and their support, Sebastian Wankerl from FAU Erlangen-Nürnberg for his great preliminary work, Christian Bergler for the inspiring exchange of ideas, Kazuki Irie from RWTH Aachen for his insights into language modeling with *rwthlm* and finally my supervisor Elmar Nöth for his unlimited amount of ideas and motivation.

Contents

1	Introduction	3
2	Medical and Theoretical Background	7
2.1	Dementia	7
2.1.1	Forms of Dementia	7
2.1.2	Diagnosis – Mini-Mental State Examination	9
2.1.3	Cookie Theft Picture Description	9
2.1.4	Dementia in Speech and Language	9
2.2	Automatic Speech Recognition	12
2.2.1	General Composition of an Automatic Speech Recognition System	12
2.2.2	Speech Recognition System Evaluation	13
2.3	Deep Learning	14
2.3.1	General Concept	14
2.3.2	Common Activation Functions	15
2.3.3	Optimizing a Network	16
2.3.4	Popular Types of Layers	16
2.4	Language Models	17
2.4.1	N-gram Language Models	18
2.4.2	Language Model Interpolation	19
2.4.3	Deep Neural Network Language Models	20
2.4.4	Language Model Evaluation – Perplexity	21
3	Materials and Methods	23
3.1	Recordings from University Clinic Erlangen	23
3.2	DementiaBank Database	23
3.2.1	DementiaBank – Pitt Corpus	24

3.2.2	DementiaBank Translations	24
3.3	European Media Lab’s Language Model Workplace	27
3.4	rwthlm – Deep Neural Network Language Modeling Toolkit	29
3.5	SRILM – N-gram Language Modeling Toolkit	29
4	Experimental Results	31
4.1	Improving Speech Recognition Performance	31
4.1.1	Linear Interpolation of a Speech Recognizer’s Language Model	31
4.1.2	Vocabulary Coverage of Test Data by Adaptation Data	33
4.2	Language Models for Dementia Detection	35
4.2.1	Experimental Setup	35
4.2.2	Experimental Results	37
4.2.3	Analysis of Results	39
4.2.4	Comparison to N-gram Language Models	41
5	Summary and Conclusions	43
	List of Figures	47
	List of Tables	49
	Bibliography	51

Abbreviations and Acronyms

LM	Language Model
LMWP	Language Model Workplace
AD	Alzheimer's Disease
FTD	Frontotemporal Dementia
MCI	Mild Cognitive Impairment
MMSE	Mini-Mental State Examination
LVCSR	Large-Vocabulary Continuous Speech Recognition
WER	Word Error Rate
WC	Word Correct Rate
ReLU	Rectified Linear Units
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
DNNLM	Deep Neural Network Language Model
EER	Equal Error Rate
FPR	False Positive Rate
TPR	True Positive Rate
AUC	Area under the Curve

Chapter 1

Introduction

World population aging has been one of the defining demographic processes of the twentieth century. As age is a main risk factor for neurodegenerative diseases, population's aging evokes numerous societal health issues. E-health, comprising diagnosis as well as therapy using electronic devices is not only convenient but urgently necessary to tackle these issues at the same scale they occur. Tools that reliably detect neurodegenerative diseases, first of all, could help assessing their epidemiological character comprehensively. Such tools would therefore relief physicians' workloads and make regularly monitoring the cognitive capabilities simpler.

Speech can help to provide insights into the condition of a person, too. Moreover, it has the advantage that it is easy to obtain. Therefore, the analysis of spontaneous speech represents a reasonable and cost-effective way to assess cognitive capabilities. However, dementia's pathology causes people to use language in an unforeseen way which degrades automatic speech recognition performance.

This thesis presents both parts of a convenient method to detect dementia, that works on a specific picture description, the Cookie Theft picture description. The goal is to first adapt the language model (LM) of an automatic speech recognizer to a specific domain, so that it better recognizes speech from people that suffer from cognitive impairments such as dementia. This capitalizes the fact, that cognitive tests are constrained to a domain with specific vocabulary. Second, the result of the speech recognizer is a prerequisite to evaluate a person's picture description on different LMs to predict whether a person suffers from dementia or not, and to even estimate the severity of the disease.

When examining a person, tension, anxiety or discomfort or simply having to describe the same picture multiple times might change the way people behave. To avoid this kind of variances, one solution could be to assess people's cognitive state without them noticing they are tested. As

illustrated in Figure 1.1, either a robot, in an assisted-living situation, or a smart-home device could bring up a picture, even old family pictures in a conversation. Thereto, ideally not always the same picture is used. The device might ask questions about the picture, without the person noticing being in a test situation. The picture is either even labeled by humans, or ideally relatives in the case of family pictures. If this high-quality information is not available, the domain-specific texts that are required to adapt an automatic speech recognizer's LM may be assembled by either collecting descriptions of similar pictures through tools like Google's image search¹ or by automatically creating descriptions using another object recognition software. Once the test person provides a picture description, the device can then transcribe the recording, which is then tested for dementia or similar cognitive impairments. As this procedure is quick and neither tedious nor annoying, it can be repeated at any time. This could help to determine the optimal use of pharmacological treatments of dementia and to deliver most effective care.

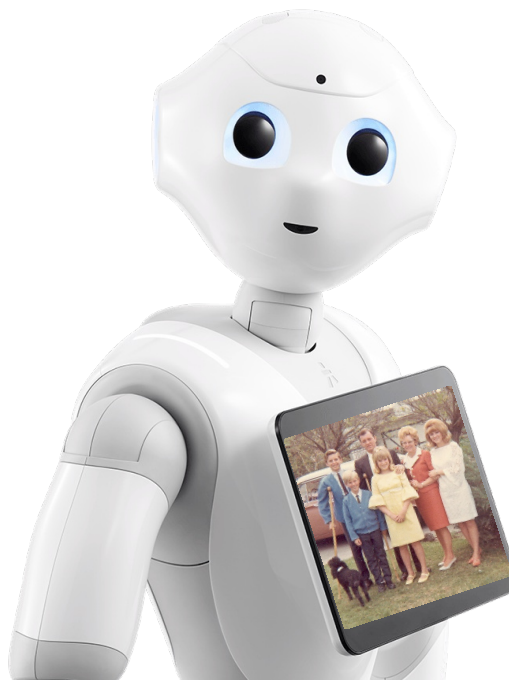


Figure 1.1: Example situation for testing a person's cognitive state²

This thesis is organized as follows: Chapter 2 gives an overview of commonly distinguished forms of dementia, its diagnosis and consequences on human speech and language. Further, the general concepts of speech recognition and deep learning are explained. Next, the basic idea of language modeling are summarized, including linear interpolation of LMs. Chapter 3 presents

¹<https://www.google.de/imghp>

²<https://www.softbank.jp/robot/special/pepper/>

the data and used tools: the DementiaBank's Pitt Corpus, its translation to German, European Media Lab's Language Model Workplace (LMWP) and the *rwthlm* deep neural network language modeling toolkit. This is followed by a report on two major experimental results: Chapter 4 begins with results on a linear interpolation of European Media Lab's base LM by domain-specific texts, namely the Pitt Corpus' translation to German. The second part is on using LMs to detect dementia by comparing the perplexities of a test set evaluated once on a model created from Alzheimer's transcription and another one created from control transcriptions. Finally, Chapter 5 draws conclusions and describes possible future work.

Chapter 2

Medical and Theoretical Background

This chapter reviews the major forms of dementia. Further, the general concepts of automatic speech recognition and deep learning are reviewed. This is followed by fundamentals of language modeling and additional background that is required to grasp the experimental results.

2.1 Dementia

2.1.1 Forms of Dementia

According to the World Alzheimer Report 2016, today 47 million people live with dementia worldwide. This number is expected to increase to more than 131 million by 2050 [Pri16]. This increase can be retraced to aging populations but also to a huge majority of people that so far have not received a diagnosis, which in consequence means they are unable to access care and treatment. That is why the rate of increase in developing countries is three to four times that in developed countries [Hus06]. The World Alzheimer Report states an urgent need to improve coverage of healthcare and lists a timely and accurate diagnosis and the optimal application of care and treatment under the top research priorities.

The term dementia describes a syndrome that is induced by neurodegenerative diseases. Numerous pathological processes affect various areas of the brain, that can lead to a form of dementia. These diseases are of chronic and progressive nature and so far are incurable. Existing medication at best allows to alleviate or decelerate symptoms. The main risk factor is age, but often neuronal damage that causes a dementia begins many years before known symptoms develop. To choose the right treatment, understanding the symptoms plays the pivotal role. The following forms of dementia are typically differentiated [Hus06]:

Alzheimer's disease (AD) accounts for around 60%¹ of all cases of dementia. It is characterized by memory loss and typically progressing to loss of cognition. At a microscopic level, neurofibrillary tangles and amyloid plaques, changes in the protein structure that cause deposits, are observed in the brain tissue. These deposits cause neurovascular dysfunctions and abnormalities of cell signaling which leads to a degeneration of neurones and loss of synapses.

Frontotemporal dementias (FTD) account for 5 to 20%¹ of all dementias. The onset of the disease typically starts early, between 45 and 65 years of age. Patients only have a mean survival of nine years from diagnosis. Males and females are concerned with equal frequency. As in early stages intellect and memory are preserved, it is often misdiagnosed. The affected frontotemporal lobe influences mood, personality and behavior, symptoms therefore include a loss of inhibitions, obsessional disorders and apathy. FTDs are usually divided into two groups: those with predominant behavioral disturbances and those with language deterioration. Language disturbances mean a reduction in verbal fluency and a loss of meaning of words. Unfortunately, no decelerating medication for FTDs is known.

Vascular dementia, which accounts for up to 20%¹ of all dementias, is characterized by a more sudden onset, usually caused by ischaemic lesions or small strokes. Risk factors are amongst others hypercholesterolaemia, hypertension, smoking, and diabetes. Additionally to other symptoms, short gait, urge incontinence and depressions are more common and severe in patients suffering from vascular dementia than in those with AD. The general course of the disease is typically shorter, with an average survival of five to seven years. That is why the main goal in treating vascular dementia is preventing any vascular events.

Lewy body dementia makes up between 15 and 20%¹ of all dementia cases. The disease seems to lie somewhere between AD and Parkinson's disease. On a microscopic level, very distinct features, so called Lewy bodies and amyloid plaques can be found. Lewy body development is accompanied by neuronal loss. Specific symptoms are hallucinations and depression, and as so far there is no medication existing to slow down Lewy body dementia, treatment focuses on improving the quality of the patient's life.

Mild cognitive impairment (MCI) is a term used to define groups of people who may be at risk of developing dementia. Cognitive impairments beyond changes expected from age and education make preventative care crucial, though insufficient to harm core activities of daily living. When memory loss is observed, the term amnesic MCI is frequently considered as a prodromal stage of AD [SO12].

Rare forms of dementia can be induced by infections, such as Creutzfeld-Jakob or HIV, or can

¹Note that the given upper limits do not sum up to 100%.

be hereditary, such as Huntington's disease [Hus06].

2.1.2 Diagnosis – Mini-Mental State Examination

As stated before, symptoms vary with the brain area affected and vary between disturbances of higher cortical functions, like memory, thinking, orientation, comprehension, calculation, learning capacity, language and judgment. Several assessment scales exist, the most common being the mini-mental state examination (MMSE). It assesses capabilities of daily self-care, like eating, dressing, walking, or looking after personal hygiene. The examination must be executed by a physician and consists of five sections: orientation, memory part 1, attention and calculation, memory part 2, language writing and drawing. The scores range from 0 to 30 points, 30 being the highest possible score. The median for healthy people with at least nine years of schooling is 29. Mild dementia is associated with a score of 21-26, moderate dementia with a score of 10–20, severe dementia with a score less than 10 [Fol75]. Figure 2.1 depicts the MMSE questionnaire.

2.1.3 Cookie Theft Picture Description

To investigate the differences in linguistic abilities, one way is the analysis of spontaneous speech. A picture description of the Cookie Theft picture, see Figure 2.2, was first included in the popular Boston Diagnostic Aphasia Examination protocol [Goo83]. The participant's task is to "describe what's going on". If the participant's description is not long enough or if he/she missed several aspects of the scene, the instructor may ask for further details. Today, the Cookie Theft picture description task is highly used and is considered a very natural approximation to spontaneous discourse. The constrained context allows good comparability to other studies using the picture descriptions, or an easy extension by more descriptions. While using computational methods, the accuracy of features usually increases as the length of a sample increases. Though, findings on the Cookie Theft picture description task can not be observed with the same intensity in other spoken language tasks [Fra16].

2.1.4 Dementia in Speech and Language

According to [Kem89], regardless of educational level, young adults produce sentences of higher syntactical complexity than do elderly adults. Furthermore stated, oral and written language of better-educated adults differs from those with less education in terms of the length of sentences they produce and the number of words they know generally. Besides and especially beyond these changes due to age, several studies have shown that all aspects of language become affected by

Mini-Mental State Examination (MMSE)

Patient's Name: _____

Date: _____

Instructions: Score one point for each correct response within each question or activity.

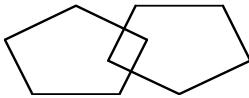
Maximum Score	Patient's Score	Questions
5		"What is the year? Season? Date? Day? Month?"
5		"Where are we now? State? County? Town/city? Hospital? Floor?"
3		The examiner names three unrelated objects clearly and slowly, then the instructor asks the patient to name all three of them. The patient's response is used for scoring. The examiner repeats them until patient learns all of them, if possible.
5		"I would like you to count backward from 100 by sevens." (93, 86, 79, 72, 65, ...) Alternative: "Spell WORLD backwards." (D-L-R-O-W)
3		"Earlier I told you the names of three things. Can you tell me what those were?"
2		Show the patient two simple objects, such as a wristwatch and a pencil, and ask the patient to name them.
1		"Repeat the phrase: 'No ifs, ands, or buts.'"
3		"Take the paper in your right hand, fold it in half, and put it on the floor." (The examiner gives the patient a piece of blank paper.)
1		"Please read this and do what it says." (Written instruction is "Close your eyes.")
1		"Make up and write a sentence about anything." (This sentence must contain a noun and a verb.)
1		"Please copy this picture." (The examiner gives the patient a blank piece of paper and asks him/her to draw the symbol below. All 10 angles must be present and two must intersect.) 
30		TOTAL

Figure 2.1: The MMSE questionnaire [Fol75]

dementia, typically starting with semantic impairment before proceeding to acoustic abnormality and syntactic impairment.

It is assumed that the cortical damage is heterogeneous in AD as the disease progresses, both within individuals but also across individuals. Consequently, linguistic symptoms should scatter just as heterogeneously. In [Fra16], 240 AD or “possible AD” samples and 233 control samples from the DementiaBank’s Pitt Corpus database [Bec94], a large variety of features on semantic, acoustic, syntactic and information impairment were collected, which are analyzed using factor analysis. The fact that the factors correlate weakly supports the hypothesis of heterogeneity. The maximum accuracy using a selection of 35 features is 81.92%. Further, among the findings are low lexical diversity, measured in Honorés statistic. A decreased proportion of nouns suggests non-specific, vague speech. It is concluded that individuals with a semantic impairment may have difficulties accessing more specific nouns, and as a result may replace these by high frequency substitutes. AD patients produced fewer verbs, fewer auxiliary verbs and fewer gerunds and participles, e. g. “he reach” is used instead of “he is reaching”, requiring an auxiliary and a participle.

Weiner et. al. [Wei08], on a database of 486 AD patients, reported a significant correlation between dementia severity and a number of different linguistic measures. More specifically, the number of animals named per minute can be used as an indicator for semantic fluency. A

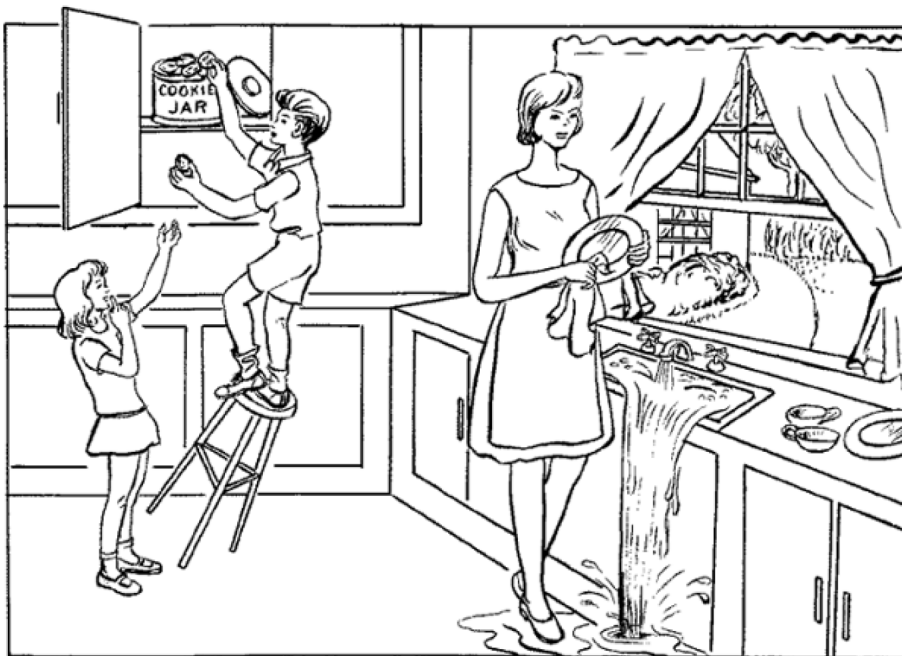


Figure 2.2: Cookie Theft Picture [Goo83]

significant association between MMSE scores and number of animals named per minute is observed. Though, abnormal articulation and repetition of words were found to be not useful for early detection of AD, but when present, are associated with a more advanced state of the disease.

Naturally, macro-linguistic language functions, such as understanding idioms, metaphors and sarcasm also tend to degrade with the progression of AD [Rap11]. However, it is stressed that these language features are worthwhile studying to evaluate abstract thinking of people affected by dementias.

2.2 Automatic Speech Recognition

2.2.1 General Composition of an Automatic Speech Recognition System

Automatic speech recognition means to map the recording of human speech to written words. The recording of human speech usually contains more information than necessary for recognition. Therefore, the goal of feature extraction is to extract the characteristic information and to remove redundant information. If X is the sequence of feature vectors of a recording and W denotes a word sequence, the most probable word sequence W^* is given by:

$$W^* = \underset{W}{\operatorname{argmax}} P(W|X) \quad (2.1)$$

After applying Bayes' theorem

$$P(W|X) = \frac{P(X|W)P(W)}{P(X)} \quad , \quad (2.2)$$

$P(X)$ can be removed, as it is independent of W :

$$W^* = \underset{W}{\operatorname{argmax}} P(X|W)P(W) \quad (2.3)$$

In order to construct a simple model which can learn from large amounts of data, the acoustic model uses feature vectors and maps them to sub-word units called phonemes. Each sub-word unit typically is represented by a sequence of feature vectors. The lexicon contains information about which words consist of which phonemes. Thus, once a sequence of phonemes is recognized, the phonemes can be assembled to a word. To further improve the automatic speech recognition system, Language Models are used. Frequent and therefore probable sequences of words are statistically modeled by assigning higher probabilities than for less frequent sequences of words, as described in Chapter 2.4.

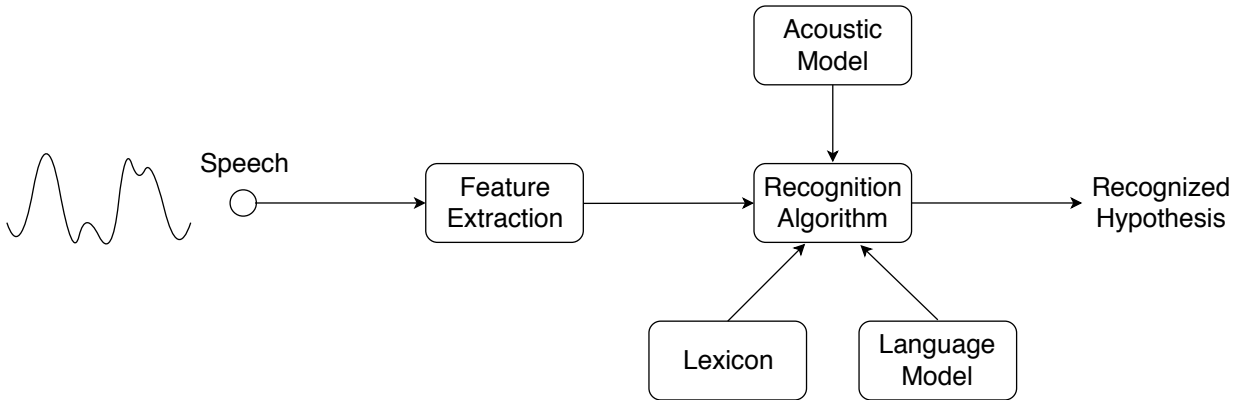


Figure 2.3: General composition of an automatic speech recognition system

Typically, the recognition system combines the acoustic model, the lexicon and the language model to a large network of probabilities. Figure 2.3 shows the major components of an automatic speech recognition system. To recognize a given speech sample, the most probable path in the network is found using so-called decoding algorithms. Sources of variation are, amongst others, the size of the lexicon or the acoustic environment. An automatic speech recognition system that is able to capture the major part of a language is often referred to as large-vocabulary continuous speech recognition (LVCSR) system [Rab08].

2.2.2 Speech Recognition System Evaluation

A speech recognition system is typically evaluated by the amount of correctly recognized words in percent. Of course, this is only possible as long as a reference is available. For N words in the reference transcript, the ASR output has three types of errors: S substitutions, D deletions, I insertions. The word error rate (WER) sums up the three types:

$$\text{WER} = 100 \frac{S + D + I}{N} [\%] \quad (2.4)$$

The percentage of correctly recognized words, or the word correct rate (WC), is given by:

$$\text{WC} = 100 - \text{WER} = \frac{N - (S + D)}{N} [\%] \quad (2.5)$$

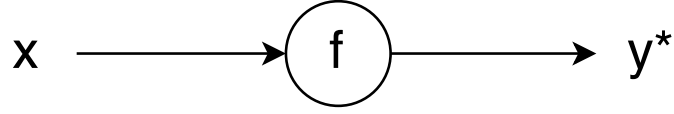


Figure 2.4: Feed-forward neural network

2.3 Deep Learning

Neural networks are a very active research field in machine learning. To let a machine learn from human experience and understand concepts, one key challenge is to provide this informal subjective knowledge to a machine. Providing a hierarchy of concepts allows the machine to learn complicated concepts by building them out of simpler ones. The provided hierarchy involves many concepts and is nowadays referred to as deep learning. The general concepts are explained in the following [Goo16].

2.3.1 General Concept

The simplest and quintessential concept of a deep learning model is the feed-forward neural network, which is a mathematical function, mapping some set of input values to output values, illustrated in Figure 2.4. The goal is to estimate a value y^* which maps an input x to a class y . The mapping $y = f(x; \theta)$ from the input by the feedforward network to the output is done by learning parameters θ . In the case of a feedforward cell, θ summarizes a weight matrix W and a bias term b :

$$y^* = f(Wx + b) \quad (2.6)$$

Neural networks come by that name, as they are typically composed to a network. Networks are assembled following two main rules: if two cells are connected in a chain, the chain rule $g(f(x))$ applies. In this chain structure, $f(x)$ is called layer 1, $g(x)$ layer 2. The overall length of the chain is the depth. The term deep learning origins from this terminology, as the depth may be large. If a cell has multiple predecessors, the inputs are added: $y = f(\sum_{i=1}^N x_i)$. Figure 2.5 shows an example 3-layer network. The first layer is usually called input layer, the last one output layer and all layers in between hidden layers, as neither output nor input of these layers are observable. The output are the estimated labels y^* . Function f is the same for all cells of one layer. Briefly, networks learn the interaction between input and output.

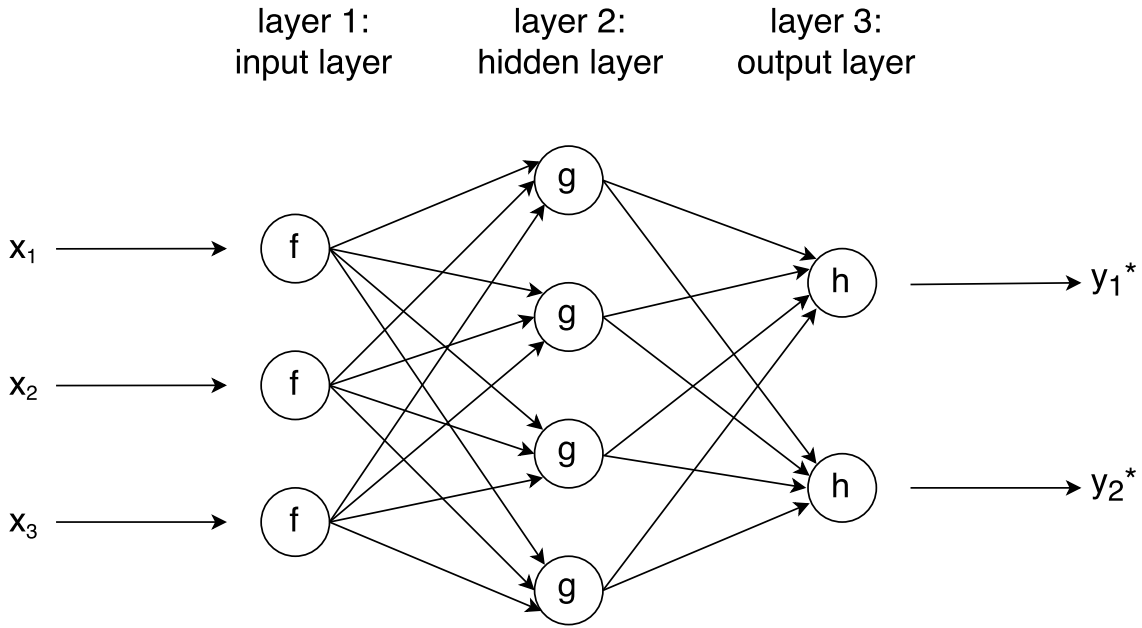


Figure 2.5: Example 3-layer feed-forward neural network

2.3.2 Common Activation Functions

To extend the mapping to represent nonlinear functions, the mapping f can be applied to a nonlinear transformation $\phi(x)$, a so-called activation function. Among the most popular ones are rectified linear units (ReLU) which use the activation function in Equation 2.7.

$$g(x) = \max\{0, x\} \quad (2.7)$$

The sigmoid function is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (2.8)$$

The tangens hyperbolicus is defined as:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (2.9)$$

The typical activation function used in output layers is the softmax function that produces a probability distribution over the output classes, thus scaling the numbers to lie between 0 and 1

and summing up to 1. It is defined as:

$$\text{softmax}(x)_i = \frac{\exp^{x_i}}{\sum_{j=0}^N \exp^{x_j}} \quad (2.10)$$

where x_i refers to the components of the vector.

2.3.3 Optimizing a Network

In order to learn the parameters θ of a network, two different modes are distinguished: forward propagation means the calculation of all outputs up to the final estimation of y^* . Backpropagation means the subsequent adaptation of the weights, so that y^* is closer to the actual y in future iterations. To measure the similarity of two probability distributions, the cross-entropy is a common measure:

$$H(y, y^*) = - \sum_i y_i \log y_i^* \quad (2.11)$$

In a machine learning context, this measure is called loss or cost function. The result is used to adapt the parameters. The respective parameters per layer are adapted according to the chain rule. In order to minimize the loss, gradient-based algorithms update the parameters iteratively:

$$\theta := \theta - \lambda \frac{\delta H}{\delta \theta} \quad (2.12)$$

where H denotes the loss function, θ denotes the network parameters and λ is the learning rate. The learning rate defines the step size of the parameter update. A training sequence that includes forward and backward propagation on all training data is typically called an epoch. Unfortunately, gradient-based optimizations have no convergence guarantee and are therefore sensitive to the choice of parameters, especially the initial values. For an overall speed-up, most parameters are summarized layer-wise, hence operations are transferred into matrix-matrix computations.

2.3.4 Popular Types of Layers

Convolutional neural networks are one special kind of neural networks that is especially well suited for data with a grid-like topology, e. g. image data or time-series data. The name indicates that the network employs a mathematical operation called convolution. A convolution is a specialized kind of linear operation, namely a general matrix multiplication with a so called kernel. Applying a kernel enhances the input or reveals structures the architecture can then learn to identify. The input and kernel are usually of the same dimension, e. g. if the input is a two-dimensional image,

the kernel would be chosen to be two-dimensional, too. Convolutional networks are therefore often used for object detection.

Another family are recurrent neural networks (RNN) which are very convenient for processing sequential data $x_1, \dots, x_t$. That means they are good at learning sequences, even of variable length. By sharing parameters across different parts of a model, they can generalize to sequence lengths seen during training. Characteristic is, that each cell output $y^{*(t)}$ is depending on the previous output $y^{*(t-1)}$, as stated in Equation 2.13:

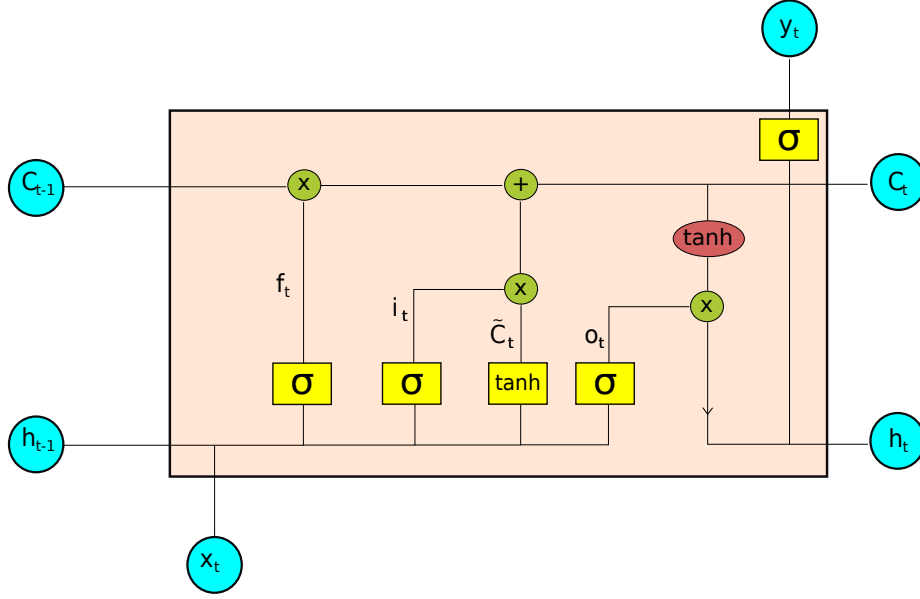
$$y^{*(t)} = f(y^{*(t-1)}, x^{(t)}, \theta) \quad (2.13)$$

When using a standard RNN, difficulties in the training process may occur, specifically the so called vanishing gradient problem, which means that the gradient when propagated back through the network either grows or decays exponentially. This requires a more sophisticated optimization algorithm. Another alternative are long short-term memory (LSTM) networks, a special kind of RNNs [Hoc97]. The concept of recurrency is expanded by introducing so called gates, to optionally let information through and remember long-term dependencies. LSTMs have a cell state C_t , a hidden state h_t , input gate i_t , forget gate f_t and an output gate o_t . All gates use a sigmoid function and output values between 0 and 1. Their values control the amount of information they pass through where 0 means nothing and 1 everything. That means basically: the forget gate controls the amount of the previous state h_{t-1} that is passed on to the cell state C_t . The input gate i_t controls the amount of the current input x_t that is passed on to the cell state C_t . The output gate also controls the amount of h_{t-1} and x_t that is passed on the hidden state h_t and the output y_t [Goo16]. Briefly, LSTMs work well, because they can accumulate information over a long duration and they provide a layout that can learn when to remember and when to remove information instead of defining a fixed context length.

2.4 Language Models

Language Models are widely used for automatic speech recognition where they are used to resolve acoustically ambiguous utterances, and on that context sometimes referred to as grammar model, but also in machine translation or information retrieval. The traditional approach is to create an n-gram LM, which has been the most successful form of LMs for a long time. However, today deep learning Language Models outperform n-grams.

²<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Figure 2.6: Long Short Term Memory cell²

2.4.1 N-gram Language Models

N-gram LMs attempt to reflect the frequency with which each word occurs in natural text. The probability for a sequence s is estimated by the product of the probabilities of the words that compose the sequence. This is simplified by only considering a history of $n - 1$ preceding words

$$P(s) = \prod_{i=1}^l P(w_i | w_1^{i-1}) \approx \prod_{i=1}^l P(w_i | w_{i-n+1}^{i-1}) \quad (2.14)$$

where w_i^j denotes words $w_i, \dots, w_j$ and n is the number of preceding words taken into account. In practice, n is often set to one, two or three, corresponding to unigram, bigram, trigram models, respectively. The bigram probability for an example sentence is put together by

$$\begin{aligned} P(\text{" . I like italian food. "}) &= P(\text{" I " | " . "}) \cdot P(\text{" like " | " I "}) \\ &\quad \cdot P(\text{" italian " | " like "}) \cdot P(\text{" food " | " italian "}) \\ &\quad \cdot P(\text{" . " | " food "}) \end{aligned} \quad (2.15)$$

whereas the trigram probability for the sentence is put together by

$$\begin{aligned}
P(\text{" . I like italian food. "}) &= P(\text{" I " | " . "}) \cdot P(\text{" like " | " . I "}) \\
&\cdot P(\text{" italian " | " I like "}) \cdot P(\text{" food " | " like italian "}) \\
&\cdot P(\text{" . " | " italian food "}) \quad .
\end{aligned} \tag{2.16}$$

Estimating the probabilities is a maximum likelihood estimation. As an example, the bi-gram formula would be:

$$P(w_i | w_{i-1}) = \frac{\text{count}(w_{i-1}w_i)}{\text{count}(w_{i-1})} \tag{2.17}$$

where $\text{count}(\ast)$ means the number of occurrences of that sequence in the training data. One of the key issues in n-gram language modeling is the sparsity of training data. Considering, for example, a text corpus with a vocabulary size of 100000 and trigram word sequences gives $100000^3 - 1 = 10^{15} - 1$ potential combinations. The probability of a sequence of words s that doesn't occur in the training data results in assigning $P(s) = 0$ where the probability clearly should be larger than zero. Zero probabilities in automatic speech recognition can lead to errors as they reject a combination, regardless of how likely the acoustic signal is. Smoothing, meaning a redistribution of the probability mass, solves that issue and adds some generalization to the model. Several smoothing techniques for n-gram LMs exist. Their performance depends on the amount of training data and history size [Che99]. In order to illustrate the concept, the add-one or Jeffrey smoothing technique means to assume each combination occurs once more than it actually did, resulting in:

$$P(w_i | w_{i-n}) = \frac{\text{count}(w_{i-n}, \dots, w_i) + 1}{\text{count}(w_{i-n}) + |V|} \tag{2.18}$$

where V is the vocabulary size of the training data. Other techniques, for example, use a backoff strategy, which means to use a lower order n-gram if an n-gram is not observed or mix different order n-grams by interpolating their probabilities.

2.4.2 Language Model Interpolation

One precondition for building n-gram LMs is that enough text data is used to estimate the model parameters. This holds true especially for higher order n-grams. However, n-gram LMs suffer from sparse data when domain-specific text data is not available or not of sufficient quality. Instead of directly adding data and re-creating a new LM, which may not improve the performance, interpolation of LMs has proven to be useful. Two approaches are distinguished: combining

counts and combining models. In the brute-force approach of interpolating counts, the new count is:

$$C_{li}(w|h) = \lambda C_{base}(w|h) + (1 - \lambda) C_{domain}(w|h) \quad (2.19)$$

where $C_{li}(w|h)$ is the linearly interpolated count of word w with history h , $C_{base}(w|h)$ the count of word w with history h in the base model, $C_{domain}(w|h)$ the count of word w with history h in the domain-specific corpus and λ the interpolation weight. Instead of interpolating counts from both text sets, one might also interpolate two LMs directly as follows:

$$P_{li}(w|h) = \lambda P_{base}(w|h) + (1 - \lambda) P_{domain}(w|h) \quad (2.20)$$

where $P_{li}(w|h)$ is the linearly interpolated probability of word w with history h , $P_{base}(w|h)$ the probability of word w with history h in the base model, $P_{domain}(w|h)$ the probability of word w with history h in the domain-specific corpus and λ the interpolation weight.

Typically, the range of the interpolation weight λ is chosen to be $0 < \lambda < 1$. According to Equation 2.19, and Equation 2.20 the lower λ , the higher the emphasis on the training corpus. For tuning the interpolation weight, relying on perplexity as metric can sometimes be misleading, as its correlation with the accuracy of a speech recognizer is not always strong [Zho02].

2.4.3 Deep Neural Network Language Models

As for n-grams, the goal of building a deep neural network language model (DNNLM) is to learn the joint probability of a sequences of words. However, DNNLMs are designed to overcome the sparse data problem by using a different representation of words. Words are encoded by vectors, which will learn to encode words with similar meaning close to each other in a continuous vector space. These word encodings are often called word embeddings. For example, the words *dog* and *cat*, both being pets map to vector space representations, which will influence predictions that will be made for sentences that contain *dog* and vice-versa. By relating words to each other and by that also sentences, a generalization forms that can tackle the sparse data problem.

The general process, explained by an example network with one hidden layer, boils down to the following steps, visualized in Figure 2.7: each word in the vocabulary is represented by an N-dimensional vector where only the index of that word is 1 and the rest of the entries are 0. This is called a one-hot encoding. A projection layer, which concatenates the words and maps them to their vector space representation is used next. Basically, this projection is a look-up table with $|V| \times P$ entries, where $|V|$ is the vocabulary size and P the dimension of the projection layer. The projection layer is followed by a variable number of hidden layers, which is followed by an

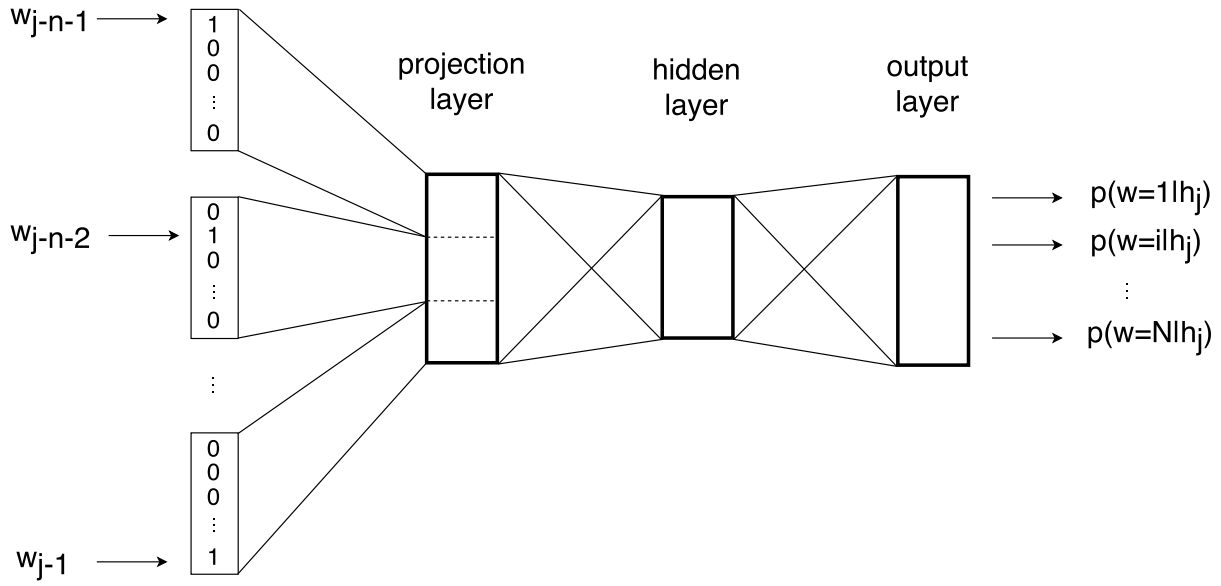


Figure 2.7: Deep neural network language model architecture [Ari12]

output layer of $|V|$ target probabilities. The output probabilities $p(w_j|h_j)$ for each word w_j , given history h_j are computed by applying the softmax function. In this case, the probability mass is distributed over only the target words, which means that unknown words are assigned zero unless a token for unknown words is introduced [Ari12]. Finally, the network's parameters are optimized according to the scheme described in Chapter 2.3.

When an architecture as depicted in Figure 2.7 is used to build a DNNLM, the following has to be considered: using feed-forward layers will, according to the chain rule, result in predicting words of all predecessors of a sequence [Sun12]. As described in Chapter 2.3, LSTMs not only address the vanishing gradient problem, but, by making use of a cell's design, e. g. hidden state and forget gate, keep or loose information over iterations. This allows for sometimes longer, but definitely variable context lengths.

2.4.4 Language Model Evaluation – Perplexity

The performance of an LM in predicting a test sample, meaning a sequence of words $S = w_1, \dots, w_N$, is often measured by calculating its perplexity:

$$PPL(S) = P(S)^{-\frac{1}{N}} \quad (2.21)$$

When evaluating an LM, a good language model tends to assign high probabilities to the test sequence S . The perplexity is therefore inversely related to the average probability a model

assigns to a sequence of words S . Note, that the perplexity is normalized by the number of words N . Lower perplexity correlates with better performance though better predictability of the test sequence [Che99]. Considering the example sequence $S_{ex} = \text{“ what is perplexity ”}$, its probability is given by:

$$P(S_{ex}) = P(\text{“perplexity”} | \text{“what”, “is”}) \cdot P(\text{“is”} | \text{“what”}) \cdot P(\text{“what”}) \quad (2.22)$$

According to Equation 2.21, the sequence’s perplexity is given by:

$$PPL(S_{ex}) = P(\text{‘What is perplexity’})^{(-\frac{1}{3})} \quad (2.23)$$

Assuming $P(S_{ex}) = 0.5$ would give $PPL(S_{ex}) \approx 1.26$. The resulting perplexity depends on the test sample, therefore perplexity is not a definite way to evaluate an LM. An LM that perfectly predicts a sequence $P(S) = 1$, would result in $PPL(S) = 1$ which is why an LM that assigns $P(S) = 0$ produces $PPL(S) = \infty$. Obviously, the value infinity should be avoided. Considering another example: An LM of vocabulary size N that predicts every word W with equal probability $P(W) = 1/N$, produces always $PPL(S) = N$. Therefore, an LM of vocabulary size $N = 10000$, that produces a perplexity on a test sequence $P(S) = 100$, has improved by factor 100.

Chapter 3

Materials and Methods

This chapter aims to give an overview of the data and tools that were used for conducting the experiments in Chapter 4. At first, the recordings from University Clinic Erlangen are presented. Next, the DementiaBank’s Pitt corpus at its translation are overviewed. Finally, the tools European Media Lab’s LMWP, *rwthlm* toolkit for building and applying DNNLMs, and SRILM toolkit for n-gram LMs are presented.

3.1 Recordings from University Clinic Erlangen

The Universitätsklinikum Erlangen provided 45 recordings from 36 German speakers of the Cookie Theft picture description. The files were recorded at 44.1 kHz and 16 bit depth stereo. All recordings were hand-transcribed by the author of this thesis by transcribing only lexically correct words, thus no colloquial language, noises or hesitating sounds. Additionally to the recordings, we received an anonymized list, which contains date of birth, gender, a diagnosis and an MMSE score to each filename. The 45 files contain speakers diagnosed with FTD, MCI and recordings from control persons. As 36 speakers provided 45 recordings, 9 speakers were examined twice. Table 3.1 shows number patients, average MMSE score and age per diagnosis ¹. Further, Figure 3.1 illustrates a histogram of all patients’ MMSE scores for both dementia and control patients ¹.

3.2 DementiaBank Database

The DementiaBank is a database of multimedia interactions to study the effects of dementia on spontaneous speech longitudinally. It contains four sub-corpora: the Holland corpus, the

¹Note that one FTD patient received a diagnosis but no MMSE score.

Table 3.1: Overview of number, MMSE score and age of 45 recordings from University Clinic Erlangen per diagnosis

	#			MMSE			Age		
	#	#♂	#♀	avg.	avg. ♂	avg. ♀	avg.	avg. ♂	avg. ♀
FTD	24	16	8	26.4	27.5	24.3	66	68	63
MCI	4	2	2	27.8	28.0	27.6	61	59	63
Control	17	10	7	29.0	29.1	28.7	70	71	69
Σ	45	28	17	27.5	28.1	26.5	67	68	65

Kempler corpus, the Pitt corpus and the PPA corpus. The corpora contain mainly English, but also German, Mandarin, Spanish and Taiwanese language and always contain audio recordings and the respective transcriptions of different tasks.¹

3.2.1 DementiaBank – Pitt Corpus

The Pitt corpus, collected between 1983 and 1988, contains recordings and transcriptions of four types of tasks: the word fluency task, the story recall task, the sentence construction task and the Cookie Theft picture description task. The data were gathered at the University of Pittsburgh School of Medicine. Participants of the study were 98 controls and 194 persons suffering from some sort of dementia, thus a total of 292 participants. The dementia diagnosed participants include a majority of AD or probable AD but also people with other dementia diagnoses. The examinations were conducted on a yearly basis up to five times, each time recording a variety of mental examinations, such as MMSE, but also age or years of education.

From the overall quantity, a selection was made for experiments in Chapter 4: 168 patients were selected, who were diagnosed with AD or probable AD, contributing 255 transcriptions. This group consists of 55 males and 113 females. From the control group, 98 speakers provided 244 transcriptions, consisting of 31 males and 67 females. In the experiments in Chapter 4, only the Cookie Theft transcriptions are used, where all additional notation was removed. Figure 3.2 shows a histogram of MMSE scores in both Alzheimer and control group.

3.2.2 DementiaBank Translations

The major prerequisite for the experiments in Chapter 4.1 is German text data of the Cookie Theft picture description domain, which is hard to get in sufficient amounts. To work around this challenge, the whole Pitt corpus was translated using three different state-of-the-art machine

¹Note, that for not all of the named languages, recordings of the Cookie Theft picture description task are available.

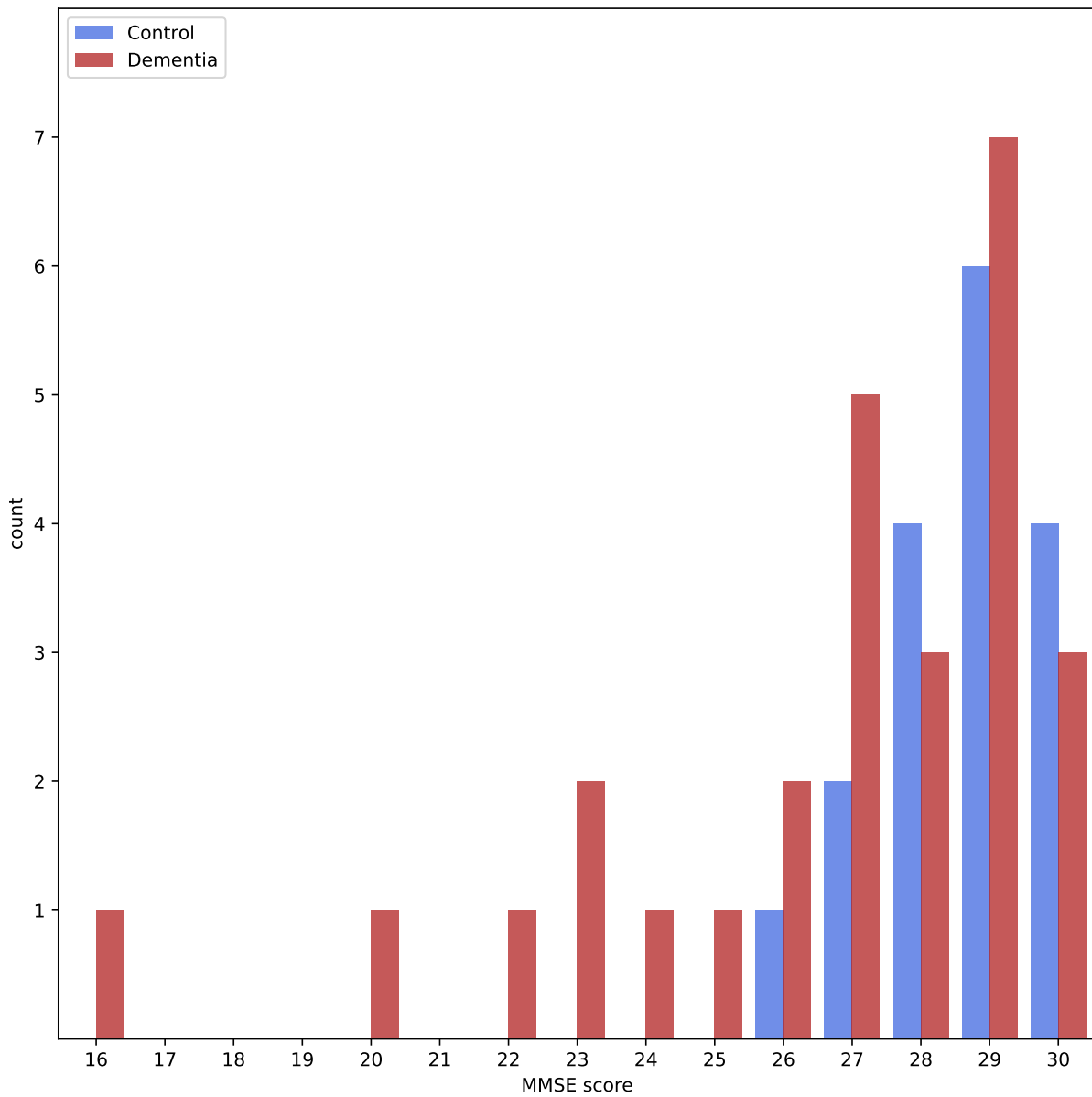


Figure 3.1: Histogram of all patients' MMSE scores for both dementia and control patients

translators: Microsoft's Bing translator², DeepL's translator³ and Google Translate⁴. Generally, the plain text files were translated, where all additional notations were removed. The texts were converted to contain only lowercase letters. Special punctuation marks were removed as well, leaving only commas, points, exclamation marks and question marks. Just as the English texts,

²<https://www.bing.com/translator>

³<https://www.deepl.com/translator>

⁴<https://translate.google.com>

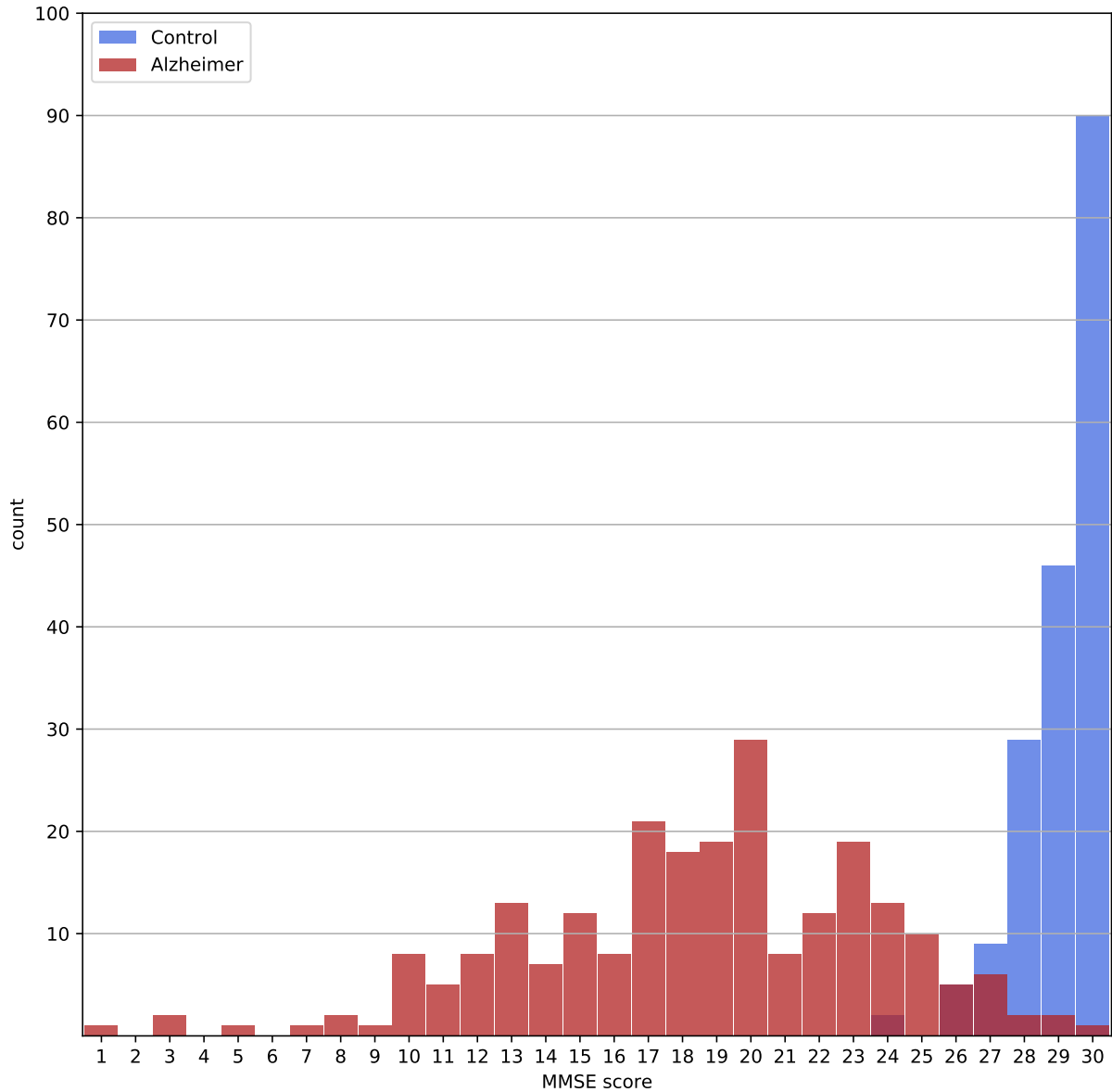


Figure 3.2: Histogram of all MMSE scores in the Pitt corpus

the German translations were converted to contain only lowercase letters. The translators differ in quality characteristics and performance, as shown in Table 3.2. Some phrases, such as “das kleine mädchen greift nach einem keks” occurs in both DeepL and Google, but not at all in Bing’s. Similarly, Microsoft’s translator consistently translates the English phrase “the dishes” by “die gerichte” while considering the context, “das geschirr” would be the correct translation. The dementia group data adds up to about 3600 sentences or 27500 words group translation, the control data to 3026 sentences or 24000 words.

Table 3.2: Most frequently occurring German translations of the control group transliterations ordered by length of phrase

phrase	occurences per translation:		
	Bing	DeepL	Google
das kleine mädchen greift nach einem keks	0	9	7
die mutter schaut aus dem fenster	7	7	7
das ist alles was ich	39	27	40
das wasser läuft über	18	42	19
das kleine mädchen	85	81	85
das geschirr	0	93	93
die gerichte	88	0	0
wasser	331	318	315

3.3 European Media Lab's Language Model Workplace

The EML European Media Laboratory GmbH located in Heidelberg maintains a general-purpose LVCSR which operates at 16 kHz. The recognizer itself can be reached via a transcription server. Besides, a platform with a graphical user interface for the development of LMs is available via another server: the so called LMWP. Both services are available in a variety of languages, amongst others American English, British English, German, Spanish and Greek.

The LMWP supports building an LM from scratch as well as the adaptation of an existing model. In addition to that, one may upload vocabulary files, pronunciation lexica, or class files, where placeholders are mapped to the corresponding words. The LMWP allows to upload text data in different formats, that can be used to create a new LM or to adapt the base LM of the language. The basic workflow of the LMWP is depicted in Figure 3.3: Generally, the LMWP

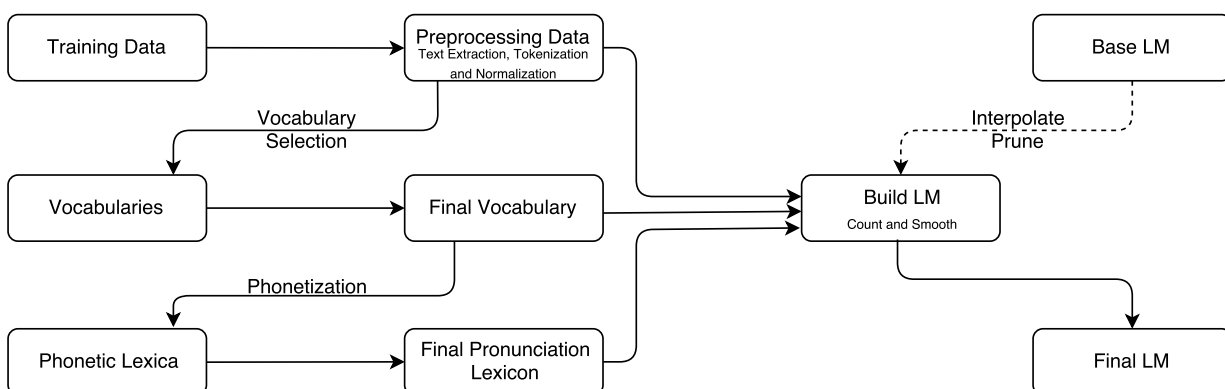


Figure 3.3: Basic LMWP workflow

uses an n-gram approach for language modeling, but note that this illustration only shows the main components as the actual algorithms constantly evolve. First, the provided training data are preprocessed. The extracted vocabulary is used to build a final vocabulary and final pronunciation lexicon. These are either used to build a new LM or to interpolate the LVCSR system's base model. In the adaptation process, many configurations can be made, the interpolation weight being the most impactful. Figure 3.4 shows LMWP's training configuration mask.

Start training quickstart_fauUser2_broadcastnews_german_16khz

Comment

☒ Generate pronunciations

Preprocessing settings

Vocabulary selection method

Minimal frequency [2]

Phonetizer alternatives

Language Model Parameters

Minimal Sentence Length

Maximum percentage unknown words per phrase

unique ☐

Automatically optimize weights ☐

Punctuation weight

Class weight

Smoothing away n-grams occurring X times or less

1grams

2grams

3grams

4grams

Weight for LM-interpolation

Minimal probability

Priority:

Figure 3.4: Screenshot of LMWP's training configuration mask

For evaluation, uploaded test data can either be just text, to rate the base or adapted model in terms of perplexity. If audio data and a corresponding text file are uploaded, the base or adapted LM are combined with the language's acoustic model, and the audio data is sent to European Media Lab's transcription server. The recognizer's results are displayed in a convenient way, including WC or a list of all out-of-vocabulary words in the test data.

3.4 **rwthlm – Deep Neural Network Language Modeling Toolkit**

The *rwthlm* toolkit is an open source C++ library that implements the concept of DNNLM and is licensed under the *RWTH ASR License* provided by the RWTH Aachen University. It allows CPU parallelization; to date, GPU usage is not supported. Internally, several speedup techniques feature a tool that can operate on large vocabulary sizes. The development was focused on the implementation of standard RNNs and especially LSTMs. The toolkit offers all features necessary to train a neural network. The training data is read in by creating one-hot encoding of the words using a supplied vocabulary file or word classes. The network architecture is customizable: Supported layer types are feedforward, standard RNNs and LSTMs. On top of that, common activation functions are available. Like many other functionalities, using a bias for the layers can be deactivated by setting a flag. Shuffling of the training samples, which is typically advised in neural network literature, can be performed sequence-wise. However, the definition of a sequence is customizable.

The networks are commonly trained with stochastic gradient descent, and the gradient is computed with the backpropagation-through-time-algorithm on a sequence of words. The parameter updates in the optimization highly depend on a good initial rate. Conceptually, faster convergence is achieved by a high initial learning rate but too high values cause the perplexity to fluctuate. This behavior is handled by decreasing the learning rate as soon as a full epoch leads to a degradation in perplexity. Briefly, a suitable initial learning rate is crucial [Sun14].

3.5 **SRILM – N-gram Language Modeling Toolkit**

The SRILM toolkit is a set of C++ libraries for building and applying statistical LMs. Released in 1995, the toolkit is still under development. It is provided by SRI International's Speech Technology and Research (STAR) Laboratory under SRI's Research Community License, which allows free usage for noncommercial purposes and runs on both UNIX and Windows systems. The main type of LMs supported are n-grams. The main functionalities to build an LM therefore are *ngram-count* to create counts and *ngram* to build an LM out of the counts. Various options are available, the most important one being the order of n-grams to use. Further, smoothing algorithms such as Good-Turing discounting, Witten-Bell, Kneser-Ney or Katz-smoothing are available as well as evaluation of test sets in terms of perplexity. Besides that, N-best rescoring, out-of-vocabulary word counting, and hit-rates of various levels of n-grams are implemented [Sto02]. More recently, the toolkit was improved by adding implementations of metacount statistics, binary file formats for more efficient reading and storage, or vocabulary subsetting, just

to name a few [Sto11].

Chapter 4

Experimental Results

This chapter contains two major parts: Chapter 4.1 summarizes the results of using text data to improve the performance of a state-of-the-art speech LVCSR system by adapting its LM. Chapter 4.2 is about using LMs, specifically the *rwthlm* toolkit, to detect whether a transliteration originates from a demented or a healthy person.

4.1 Improving Speech Recognition Performance by Adapting the Language Model

Having a speech recognizer that works well on distorted speech is the main condition for using LMs for dementia detection. Therefore, this experiment aims to improve the performance of European Media Lab's German LVCSR system by adapting its LM with domain-specific text data, namely the Pitt corpus translations of different translators. The performance is evaluated in terms of WC and perplexity, both weighted over the number of words per recording of the 45 manually transcribed recordings from University Clinic Erlangen. An initial experiment showed that the average WC testing recordings down-sampled to 8 kHz is 51.6 %. This value emphasizes the need to improve speech recognition performance. Adapting a LVCSR's acoustic model requires lots of data, therefore is not a convenient approach for the scarce-resource domain of cognitive tests.

4.1.1 Linear Interpolation of a Speech Recognizer's Language Model

This experiment was conducted with the main goal to find out, how to adapt the LM of a LVCSR system to a particular domain. This concerns two main variables: the text data and the interpolation weight. As three different translations were created, performance was first evaluated for Bing's,

DeepL’s and Google’s translation but also for the three translations together, i. e. “pooled”. Those four different text quantities were then evaluated at interpolation weights $\lambda = \{0.1, 0.2, \dots, 1.0\}$ to identify the optimal value. The emphasis on the text data increases with higher interpolation weight λ . Consequently, $\lambda = 1.0$ means that the LM is only made of the new text data. Figure 4.1 summarizes these results. The baseline WC of the 45 files of the LVCSSR system is at 61.9 %, the baseline perplexity of the transcriptions evaluated on the baseline system’s LM at 576. The best WC was found at an interpolation weight $\lambda = 0.7$, where the WC reached 69.7 % and perplexity a value of 229, when “pooling” the three translations.

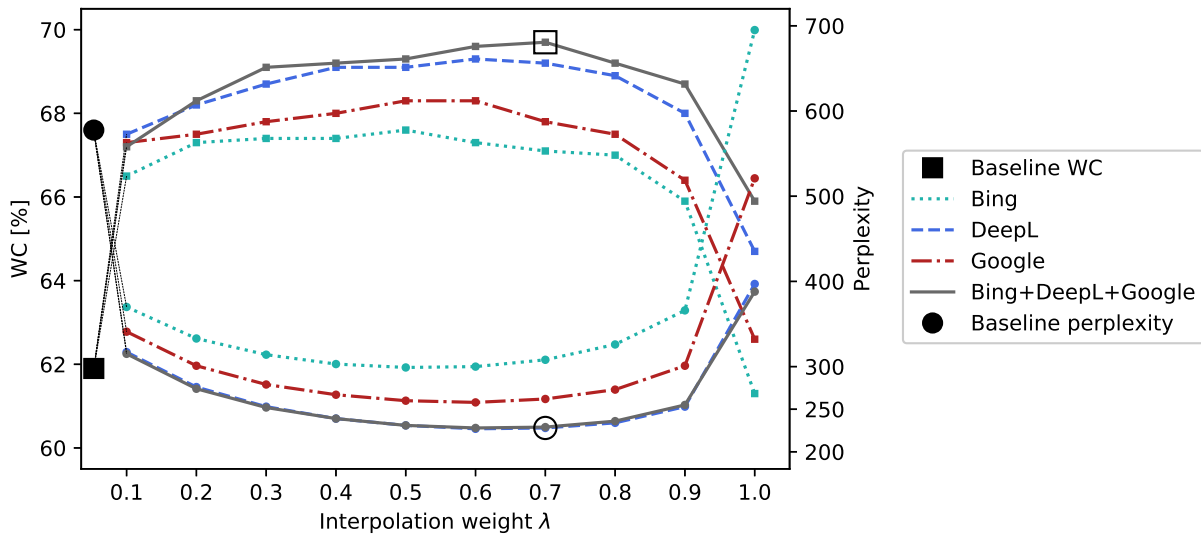


Figure 4.1: Mean WC and perplexity of the baseline LCVSR system and at interpolation weights $\lambda = \{0.1, 0.2, \dots, 1.0\}$

In order to use the available data economically, additionally, the 45 transcriptions were partitioned into nine folds, to perform a 9-fold cross-validation on the data. The files are divided into nine folds, as nine speakers provided two recordings giving five files per fold. Speakers that provided two recordings are always in the same fold. Furthermore, the files were partitioned so that each fold contains at least one male and one female speaker, at least one control speaker and no more than three speakers diagnosed with FTD.

Having found an optimal interpolation weight at $\lambda = 0.7$, Figure 4.2 illustrates the succeeding course of the experiment. The baseline result of 61.9 % WC can be improved by interpolating the baseline LM either with a large amount of poorer quality translated text data achieving 69.7 % WC or in a cross-validation setup with the manual transcriptions, which are of high quality, achieving 69.9 % WC. For this purpose, each fold is tested on an adaptation by a combination of

the three translations and the transcriptions of the remaining eight folds. Hence, the overall best result was achieved when combining manual transcription and the translations by Bing's, DeepL's and Google's translations, resulting in a WC of 71.1 % at a perplexity of 189. Naturally, as WC increases, perplexity values decrease. This can be regarded as a very solid result, considering the following: There are thirteen recordings recognized with less than 60 % WC by eleven dementia and two healthy patients. Their average WC of 45.6 % indicates that they are at a progressed stage of dementia, their average MMSE score is 25.1. In contrast to that, there are sixteen recordings recognized at more than 80 % WC at an average of 85.5 %, provided by 9 control and 7 dementia patients, their average MMSE score is 28.8. Hence, the adaptation not only improves accuracy of the control speakers.

A final experiment was conducted to give higher emphasis on the manual transcriptions: The linear interpolation of the baseline LM and the three translations at $\lambda_1 = 0.7$ was selected as a new baseline LM. A second linear interpolation of that new baseline model with the manual transcriptions resulted in a WC of 70.5 % and perplexity of 162 at $\lambda_2 = 0.6$.

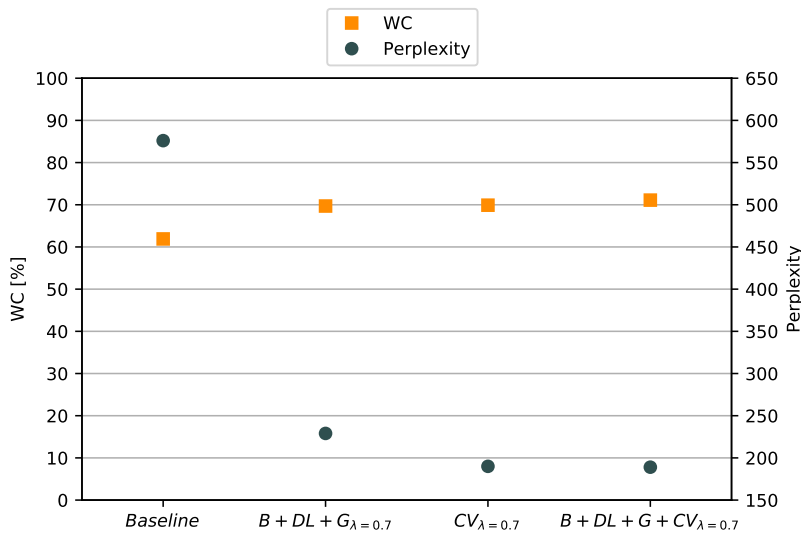


Figure 4.2: Comparison of WC and test data perplexity of the baseline LVCSR system, the adaptation in the 9-fold cross-validation scenario, the adaptation by all three translations and the combination of 9-fold cross-validation and all three translations

4.1.2 Vocabulary Coverage of Test Data by Adaptation Data

In order to better understand why certain text quantities produced better results than others, the vocabulary coverage of the 45 test transcriptions by the six different quantities was measured

and plotted in histograms. Figure 4.3 visualizes that the DeepL translations, as well as the three translations together, cover a lot more files above 90% than the Bing and the Google translations, which highly supports the results in Figure 4.1. The same holds for the cross-validation text quantity and for the three translations plus the cross-validation text data, which ranges almost entirely above 90%.

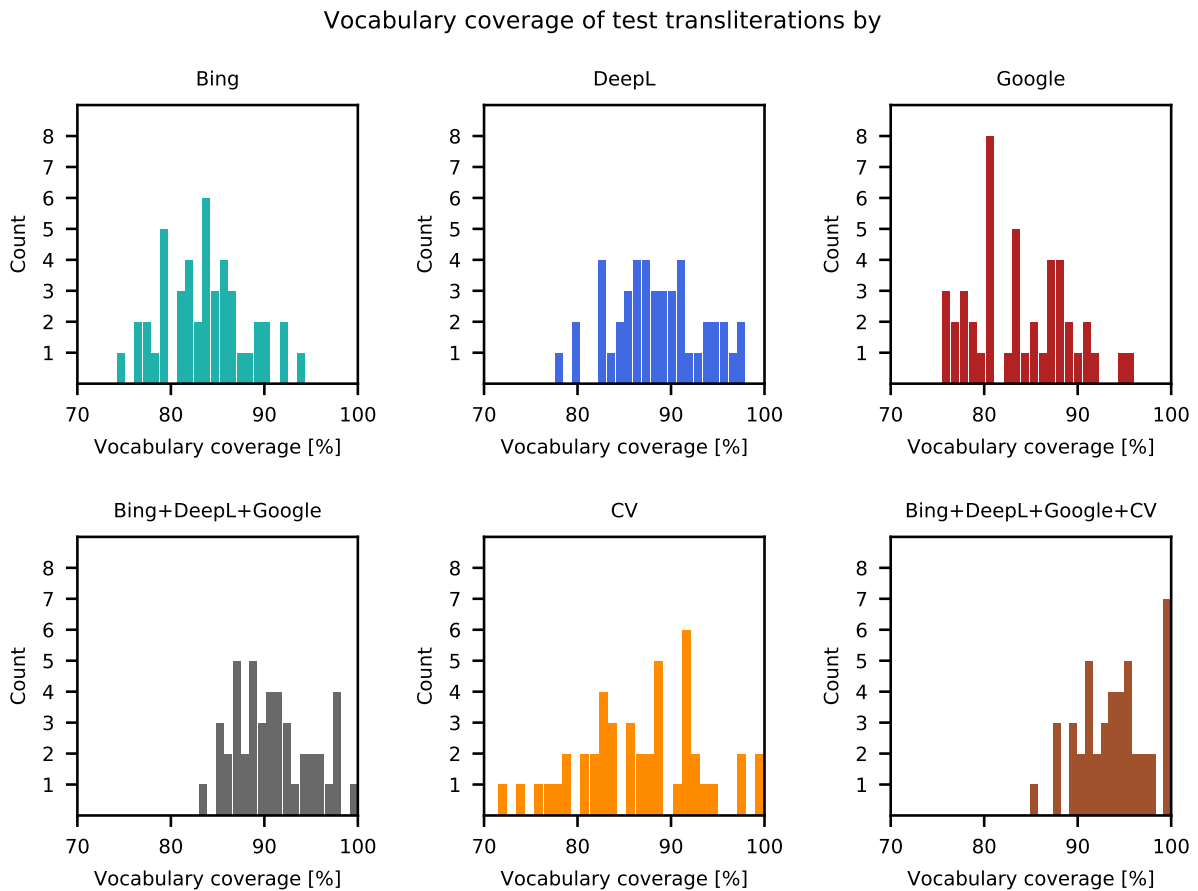


Figure 4.3: Vocabulary coverage histogram of 45 test files per text quantity

Very similar observations can be made when re-arranging the histogram to represent the partitioning into 9 folds. Bing’s and Google’s translations show a likewise low coverage of vocabulary over all folds, whereas DeepL’s translation as well as the cross-validation covers the test data’s vocabulary to a higher extent. In the end, the combination of all data yields the highest average coverage value.

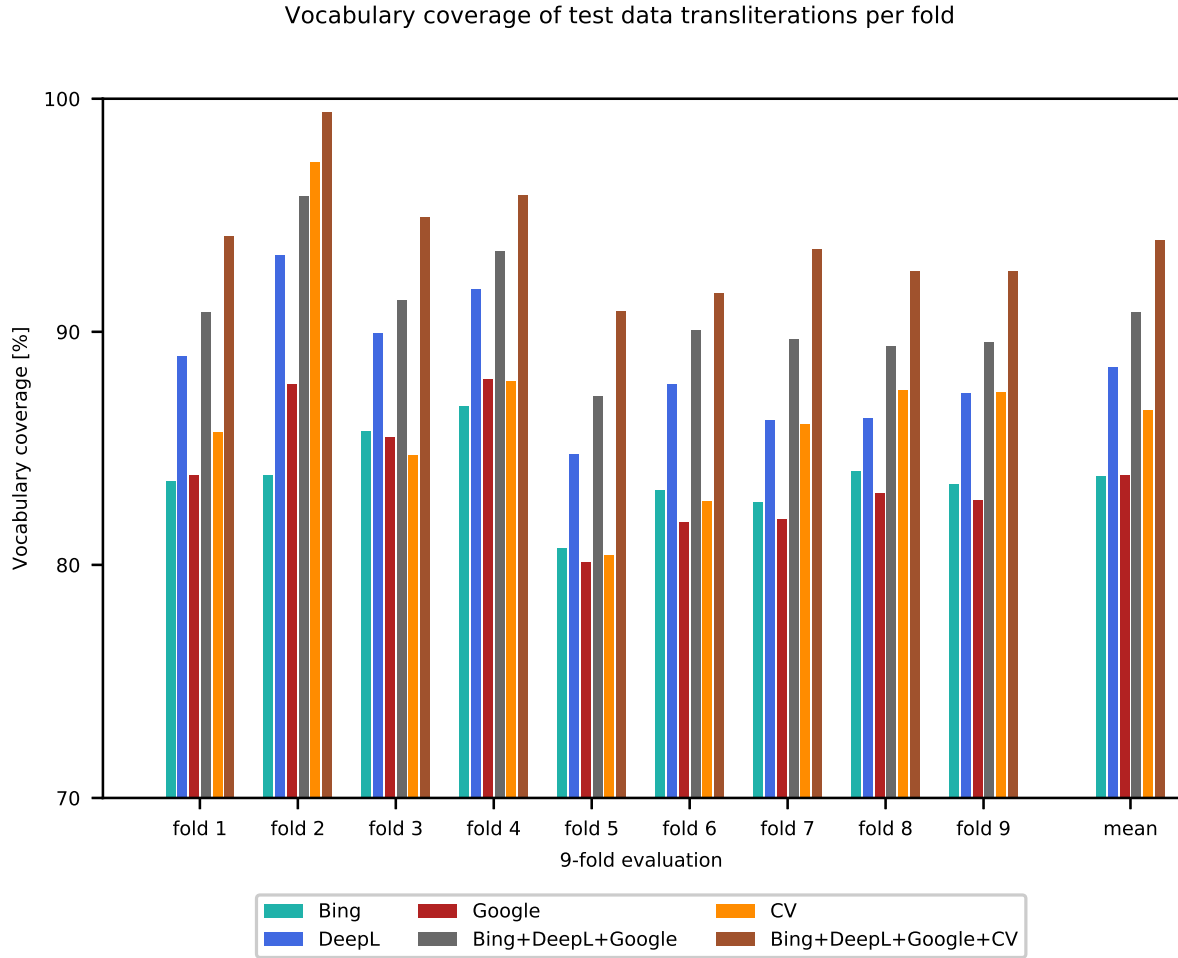


Figure 4.4: Vocabulary coverage of 45 test files by text quantity per fold

4.2 Language Models for Dementia Detection

The goal of the experiment described in the following is to classify Alzheimer’s patients by comparing the test set perplexities of their Cookie Theft picture description transliterations when evaluated on an LM of the control group’s text data and Alzheimer group text data.

4.2.1 Experimental Setup

In the following experiment, only the selection of 168 Alzheimer and 98 control participants of the Pitt corpus is used. For both groups, all transliterations per group are used to create an Alzheimer’s LM $\mathcal{M}_{Alzheimer}$ and a control LM $\mathcal{M}_{Control}$ by using *rwthlm* toolkit. These are supposed to represent the typical Alzheimer’s and healthy speech. Then, a leave-one-speaker-out

cross-validation is performed, which means that for all speakers, an LM $\mathcal{M}_{-s}$ is created, using all transliterations of that group but those of one speaker. The cross-validation is conducted per speaker, as speakers that provided multiple recordings are expected to use similar phrases that are not always representative for the whole group and would distort the results. The transliterations provided by speaker f are then used to evaluate a test set perplexity, also by using the *rwthlm* toolkit. A speaker s is always tested on $\mathcal{M}_{-s}$, giving a perplexity ppl_{own} . Additionally, the speaker is tested on the respective other LM, giving ppl_{other} using $\mathcal{M}_{Alzheimer}$, if belonging to the control group, or $\mathcal{M}_{Control}$, if belonging to the Alzheimer's group. Naturally, ppl_{own} is expected to be lower than ppl_{other} , as the own group's LM is supposed to represent the test set better than the other group's LM. Finally, the difference between the perplexity evaluated on an Alzheimer's LM and a control LM is calculated for each speaker, giving following distinction for the groups:

$$ppl_{diff} = \begin{cases} ppl_{own} - ppl_{other} & \text{if } s \in \text{Alzheimer's group} \\ ppl_{other} - ppl_{own} & \text{if } s \in \text{Control's group} \end{cases} \quad (4.1)$$

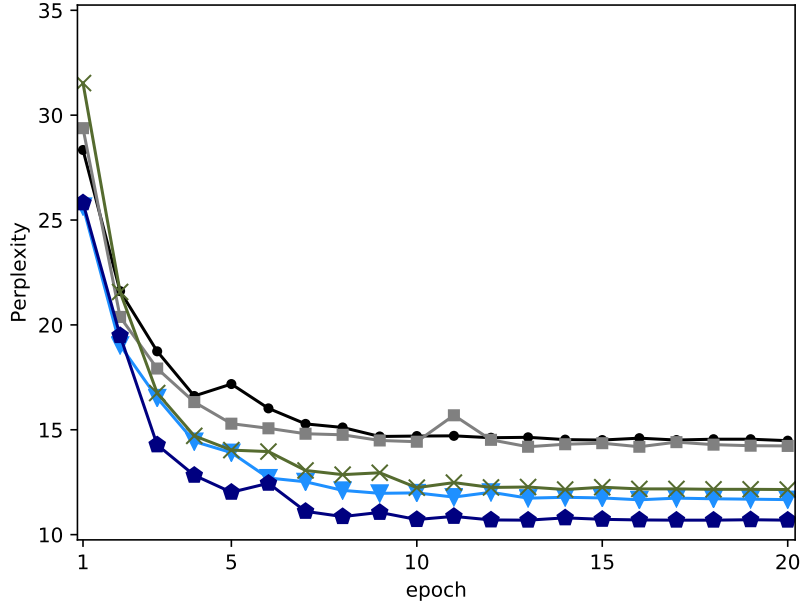


Figure 4.5: Development perplexity per number of epochs for five randomly selected model creations

In order to actually classify the transliterations, the perplexity difference ppl_{diff} is considered. A binary decision is made by setting a threshold, such that both groups have equal error rate (EER). The results are assessed at EER, as this gives a robust estimate of a classifiers performance.

Basically, all speakers are simply evaluated on an Alzheimer's LM and a control's LM, the only difference being that this is done in a leave-one-speaker-out cross-validation. The LMs are created using the *rwthlm* toolkit, described in Chapter 3.4. The tool expects a vocabulary list, a file containing the training text data, a file containing the development text data and an initial learning rate. The learning rate was tuned individually for every network architecture, so that the development perplexity after the first epoch is minimal. The development set was created by randomly selecting ten speakers of the respective group who were then excluded from the training data. Additionally, instead of applying some early stopping criterion to either learning rate or development perplexity per epoch, the maximal number of epochs is simply set to 20. As illustrated in Figure 4.5, the latest after 20, the development perplexity, here for five randomly selected model creations, flattens out.

4.2.2 Experimental Results

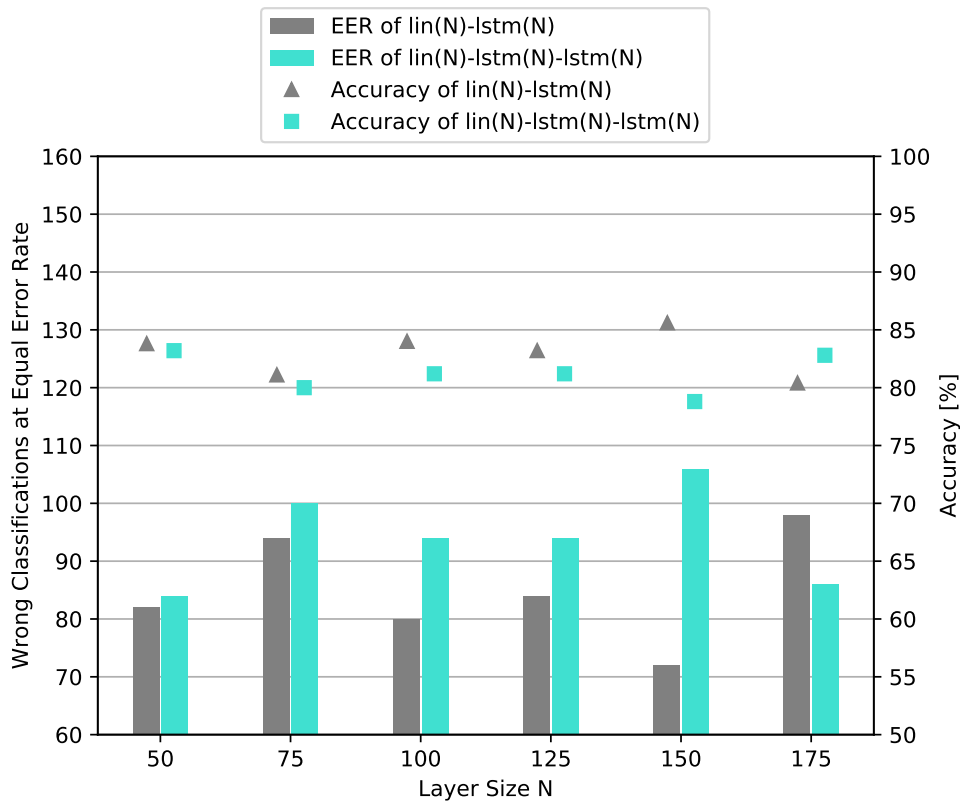


Figure 4.6: Overall result of the experiment at different architectures

The experimental setup, as described in the previous chapter can be executed for different

neural network architectures. As defined by the toolkit, the first layer must be a linear layer with identity activation function. The main focus of this experiment was on using LSTM layers, as the recurrency property allows for flexible context lengths. The results are assessed at EER. Additionally, the accuracy in percent is considered at EER, too. Figure 4.6 shows the results for different layer sizes using either one or two LSTM layers. Using only one LSTM layer seems to give a more stable result. The best result was achieved at a layer size of 150, resulting in 72 wrong classified transliterations, which means 85.6 % accuracy. Comparing the accuracy of different architectures means implicitly optimizing on the test set. This is done for lack of enough data. Besides, the step size of the layer size N was only increased in steps of 25.

Figure 4.7 shows the mean perplexities of the result using one LSTM layer of size 150. As expected, $\overline{ppl}_{own}$ is consistently lower than $\overline{ppl}_{other}$ for both groups. The threshold for separating the two groups was set to $\overline{ppl}_{diff} = -0.78$.

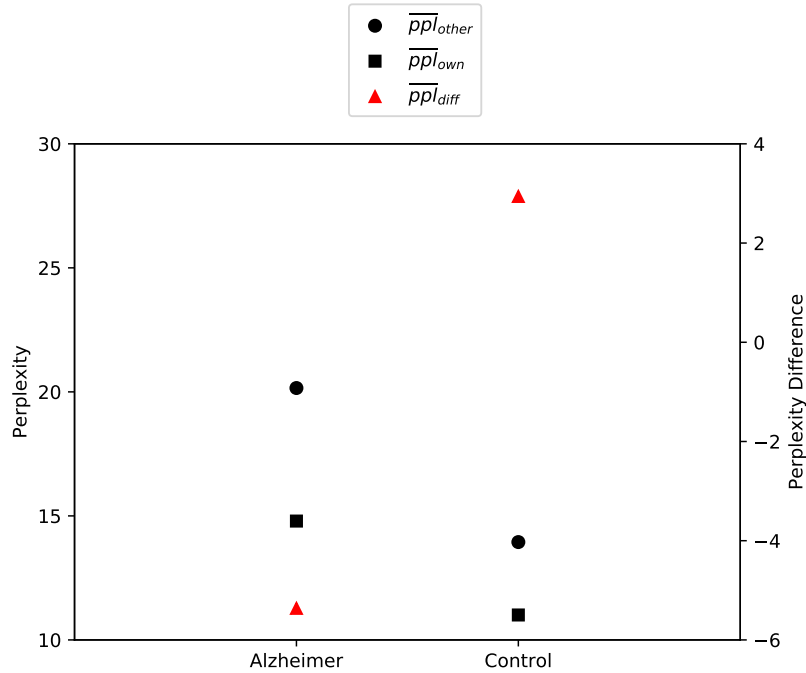


Figure 4.7: Mean values of perplexity of Alzheimer and control group

In [Rah18], the same selection of files was examined. The author of this thesis provided stemmed data. Stemming means the reduction of words to their word stem or base form. Stemming therefore is the removal of syntactic information, more precisely it results in a reduction of the number of types (the number of different words), consequently in a changed type-token ratio. Figure 4.8 shows a comparison of unstemmed and stemmed results for different layer sizes N

using one LSTM layer. The overall best result was still achieved on unstemmed data. Though, stemming seems to stabilize the results. The best accuracy at $N = 100$ gives 76 wrong classified transliterations and an accuracy of 84.6%.

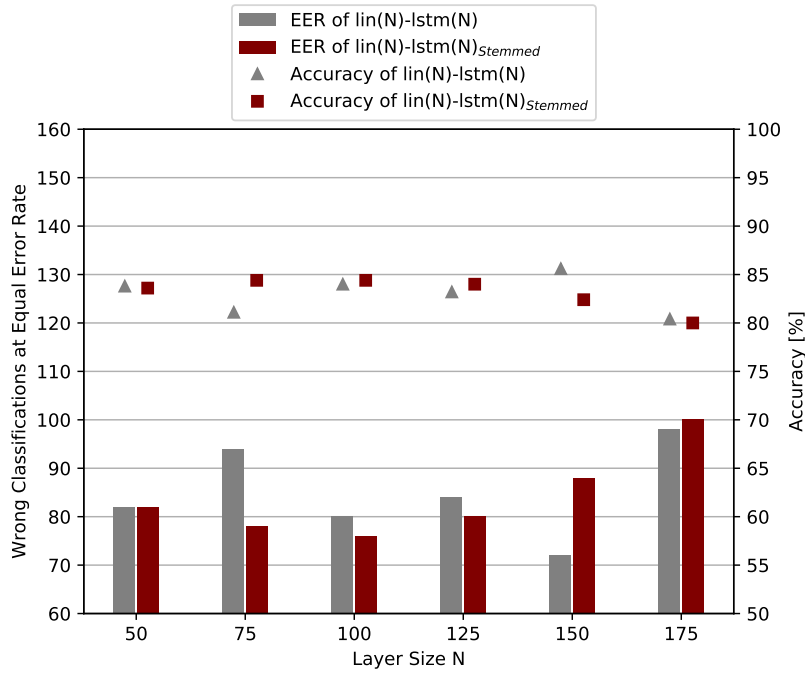


Figure 4.8: Comparison different architectures on original and stemmed data.

4.2.3 Analysis of Results

To generally grade a two-class separation, ROC curves are common, if only interested in one class. Figure 4.9 shows the ROC curve for the result using one LSTM layer of size 150: At 6% false positive rate (FPR), the true positive rate (TPR) is already at 80%, meaning that while only classifying 6% of controls wrongly, 80% of Alzheimer speakers are classified correctly. At 10% FPR, the TPR goes up to 85%, continuing to 90% TPR at 25% FPR, and at 50% FPR, the TPR is already at 95%. The area under the curve (AUC) is 0.92.

Figure 4.10 shows the histogram of MMSE scores for our selection of speakers from the Pitt corpus, for all speakers from whom an MMSE score is available. Moreover, the number of wrongly classified transliterations per MMSE score is indicated by solid red and blue lines, as well as the accuracy in percent per MMSE score indicated by dashed lines. The solid gray line indicates the overall accuracy per MMSE score. The 36 wrong classified transliterations per group in both cases come from 27 different speakers.

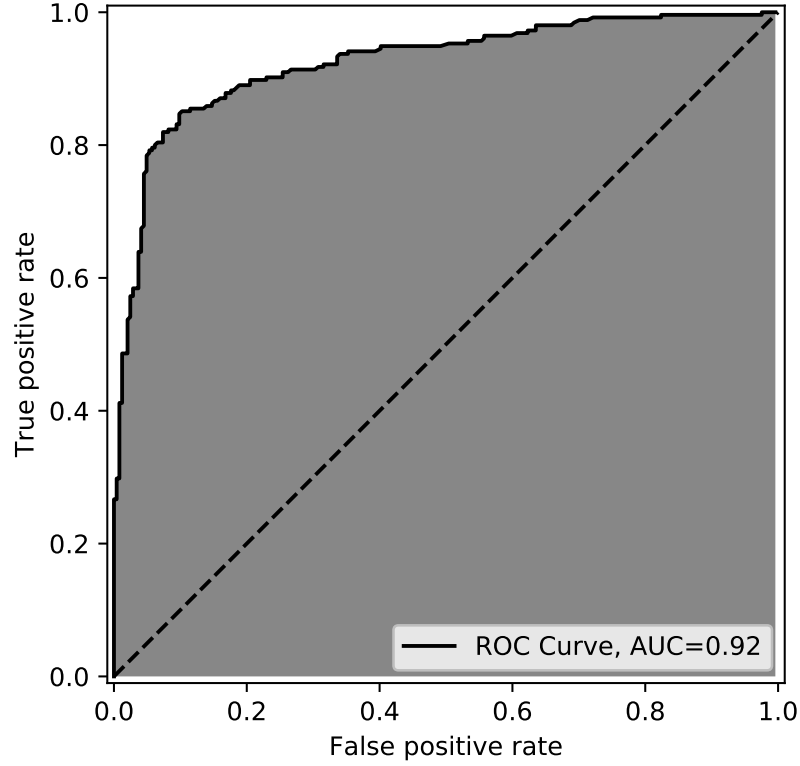


Figure 4.9: ROC curve of the result using a linear and an LSTM layer of size 150 for the binary classifier using ppl_{diff} as threshold

Table 4.1 shows Pearson’s and Spearman’s correlation between MMSE scores and the perplexity difference values ppl_{diff} . The correlation values for the Alzheimer’s group indicate that ppl_{diff} can be used to estimate a person’s MMSE score. The control group’s correlation is low. Considering, that the MMSE scores only range between 24 and 30, but the perplexity difference values still vary to a larger extend, that value is tolerable. Moreover the correlation of all MMSE values and all respective perplexity difference is promising.

Table 4.1: Correlations between MMSE scores and perplexity difference ppl_{diff}

	Correlation r	Correlation ρ
Alzheimer’s	0.433	0.547
Control	0.112	0.109
All	0.656	0.771

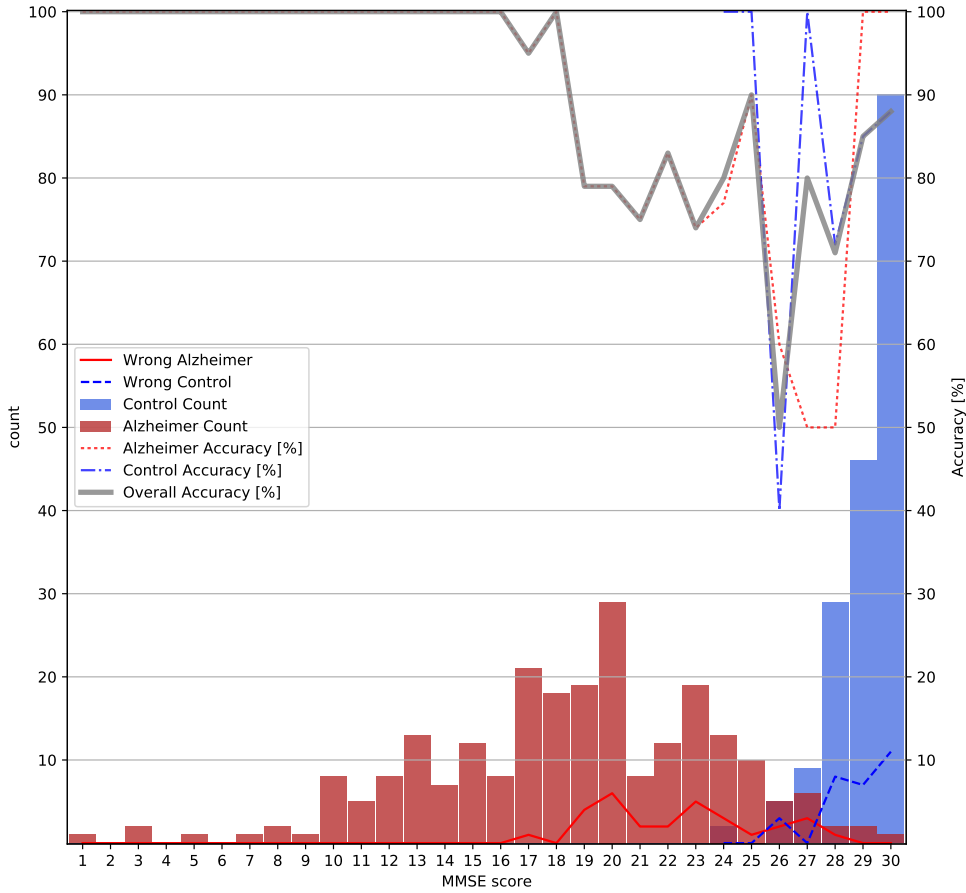


Figure 4.10: Histogram of all MMSE scores in the Pitt corpus, errors, and accuracy per MMSE

4.2.4 Comparison to N-gram Language Models

To rate the above presented results, Figure 4.11 shows the mean perplexities for the same experimental setup on the Pitt corpus selection, once using the SRILM toolkit [Sto02] as in [Wan17] and once for the *rwthlm* toolkit. The absolute values of ppl_{own} and ppl_{other} by SRILM are higher than by using *rwthlm*, though both ppl_{diff} values differ not that much, thus the discriminative power is similar. The overall accuracy when using SRILM is 77.1% at 114 wrongly classified transliterations. However, using *rwthlm* outperforms this result by 8.5% absolute or 42 wrong classified transliterations less.

The performance comparison of the two language modeling tools can also be made by comparing ROC curves: Figure 4.12 shows both curves. Using *rwthlm*, at 10% FPR, the TPR is already at 85%, compared to only at 60% using SRILM. While at 90% TPR with *rwthlm* the FPR stays at a low 25%, with, it is already at 53%. The AUC using SRILM is 0.83 compared to 0.92 achieved when using the *rwthlm* toolkit.

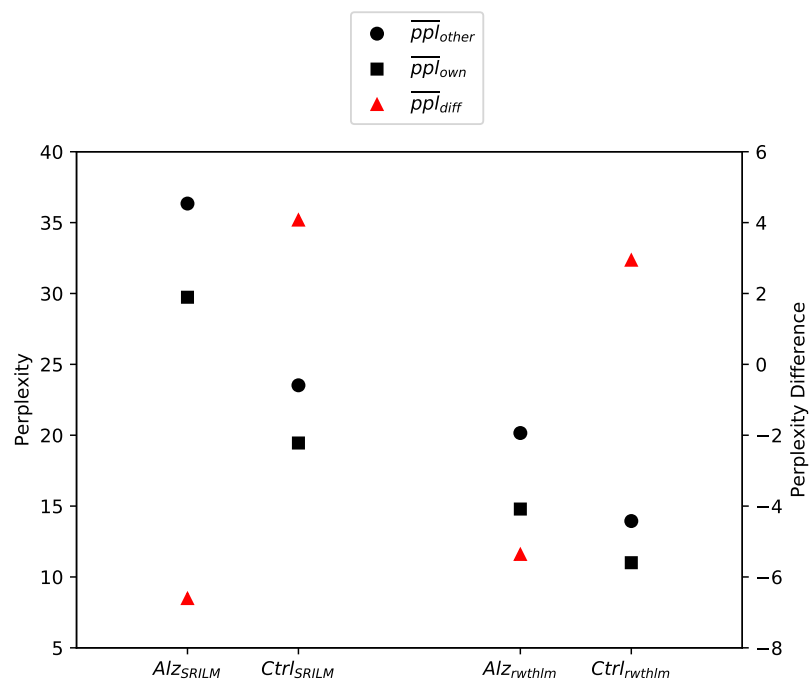


Figure 4.11: Comparison of perplexity means of language modeling toolkits SRILM and *rwthlm*

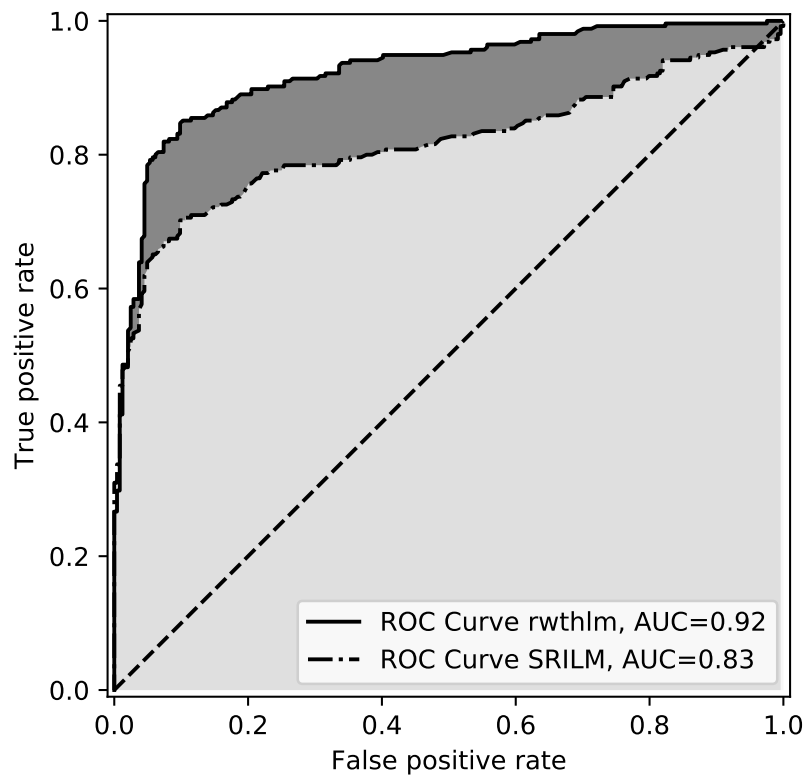


Figure 4.12: Comparison of ROC curves of the results using *rwthlm* and SRILM

Chapter 5

Summary and Conclusions

This thesis presents two approaches using language models in the context of automatic evaluation of dementia. Chapter 2 clarifies the epidemiological character of dementias, together with a summary of the forms and pathological effects of the disease. This is followed by fundamentals of automatic speech recognition, deep learning and language models. The latter implemented in the following tools in Chapter 3: The LMWP, a server for adapting and creating LMs as well as testing a LVCSR system’s LM, the *rwthlm* toolkit for DNNLMs as well as the SRILM toolkit for n-gram LMs. Furthermore, Chapter 3 contains a description of a small database of 45 recordings from the University Clinic Erlangen, the DementiaBank’s Pitt corpus and its translation to German to conduct experiments.

Chapter 4 presents two main experiments: First, Chapter 4.1 shows that an off-the-shelf LVCSR’s LM is not prepared for a domain as particular as the Cookie Theft picture. The average WC on 45 recordings at 8 kHz is only 51.6 %, and 61.9% at 16 kHz. Unforeseen usage of language by dementia patients combined with inaccurate articulation results in bad intelligibility for both humans and speech recognizers. The baseline recognition rate at 8 kHz indicates that improving a recognizer’s acoustic model may be beneficial, but would require enough data and is not offered by EML. To improve speech recognition performance, EML’s LVCSR’s LM is adapted to the domain of the examination. This domain data is a scarce resource and not available in every language. Using state-of-the-art machine translations of native English transliterations to linearly interpolate a LVCSR’s LM improved the average recognition rate to 69.7 % WC. In order to add more variation to the data, three different machine translation softwares were used. Adding only a small amount of higher-quality data of manual German transcriptions improved to 71.1 % WC at an interpolation weight $\lambda = 0.7$, which is an absolute improvement of 9.4% WC. Even though this experiment was conducted in German, one big advantage of this method is that the Pitt corpus

could be machine translated to any other language with similar outcomes. Naturally, the higher the vocabulary coverage of the respective domain (e. g. the test data) by the adaptation data, the more an adaptation is beneficial. This applies not only to the Cookie Theft picture domain but to any constrained domain. Consequently, a similar adaptation can be done to any domain of cognitive tests, e. g. also category naming tasks. The final goal may be to provide a battery of LVCSRs adapted to the domains of cognitive examinations, which would represent a major step towards fully automating dementia diagnosis. Adapting an LM is a more convenient way to tune a LVCSR to a particular domain than improving its acoustic model.

The resulting transliteration can then be used to evaluate on LMs created from either Alzheimer text data or control patient's data, as described in Chapter 4.2: Just by comparing perplexity values, the database can be separated with an accuracy of 85.6% on a total of 499 test persons. Using DNNLMs with LSTM cells improves the approach using n-gram LMs by 8.5% absolute corresponding to 42 wrong classified transliterations less. This approach exploits the assumption that Alzheimer's patients describe the picture in an unforeseen way, which leads to more unpredictable language structures resulting in higher perplexity values. It was shown that clear cases, e.g. those with low and high MMSE scores, have a high recognition rate. The borderline cases at the transition from dementia to control group having an MMSE score between 24 and 27 are less obvious and therefore have a lower recognition rate. Coincidental, correlating all MMSE scores with the respective perplexity difference yields a Pearson's correlation of $r = 0.656$ and Spearman's correlation of $\rho = 0.771$. That shows that this method can not only be used for detecting Alzheimer's disease, but can also be used to estimate a patient's MMSE score.

Thus, this thesis provides a scientific foundation for both, i. e. adapting a LVCSR system's LM to achieve better accuracy in a challenging situation, and LMs can be used to detect dementia. At the same time, traditional n-gram models were shown to be outperformed by DNNLMs. The clear advantage of this purely statistical approach of using LMs for dementia detection is that it is not restricted to the English language; it will work in another language, too. Though, to use this approach in practice, it has to be examined, whether recognition results work as good as manual transliterations.

Future research to improve this method could focus on further optimizing the neural networks architecture by using more sophisticated types of layers and using more data. Besides using a plain language modeling approach, a more advanced neural network, combining the use of LMs with other features such as gender, age, or years of education, could approach this challenge more holistically. Naturally, acoustic features are worthwhile, but require a sufficient amount of data.

Finally, this method could be extended to not only work on a specific picture description,

but use pictures related to a person's life that trigger his/her desire to talk about them. A speech recognizer can be tuned for a particular domain by collecting similar pictures using object recognition software. The recognizer's output can then be used to estimate a person's cognitive capabilities using linguistic and statistical methods, such as presented in this thesis.

List of Figures

1.1	Example situation for testing a person's cognitive state	4
2.1	The MMSE questionnaire	10
2.2	Cookie Theft Picture	11
2.3	General composition of an automatic speech recognition system	13
2.4	Feed-forward neural network	14
2.5	Example 3-layer feed-forward neural network	15
2.6	Long Short Term Memory cell	18
2.7	Deep neural network language model architecture	21
3.1	Histogram of all patients' MMSE scores	25
3.2	Histogram of all MMSE scores in the Pitt corpus	26
3.3	Basic LMWP workflow	27
3.4	Screenshot of LMWP's training configuration mask	28
4.1	Mean WC and perplexity	32
4.2	Comparison of WC and test data perplexity	33
4.3	Vocabulary coverage histogram of 45 test files per text quantity	34
4.4	Vocabulary coverage of 45 test files by text quantity per fold	35
4.5	Development perplexity per number of epochs	36
4.6	Overall result of the experiment at different architectures	37
4.7	Mean values of perplexity of Alzheimer and control group	38
4.8	Comparison different architectures on original and stemmed data.	39
4.9	ROC curve of the result using a linear and an LSTM layer	40
4.10	Histogram of all MMSE scores, errors, and accuracy per MMSE	41
4.11	Comparison of perplexity means	42
4.12	Comparison of ROC curves of the results using <i>rwthlm</i> and SRILM	42

List of Tables

- 3.1 Overview of number, MMSE score and age of 45 recordings 24
- 3.2 Most frequently occurring German translations 27
- 4.1 Correlations between MMSE scores and perplexity difference ppl_{diff} 40

Bibliography

- [Ari12] E. Arisoy, T. N. Sainath, B. Kingsbury, and B. Ramabhadran. Deep neural network language models. In *Proceedings of the NAACL-HLT 2012 Workshop: Will We Ever Really Replace the N-gram Model? On the Future of Language Modeling for HLT*, pages 20–28. Association for Computational Linguistics, 2012.
- [Bec94] J. T. Becker, F. Boiler, O. L. Lopez, J. Saxton, and K. L. McGonigle. The natural history of Alzheimer’s disease: description of study cohort and accuracy of diagnosis. *Archives of Neurology*, 51(6):585–594, 1994.
- [Che99] S. F. Chen and J. Goodman. An empirical study of smoothing techniques for language modeling. *Computer Speech & Language*, 13(4):359–394, 1999.
- [Fol75] M. F. Folstein, S. E. Folstein, and P. R. McHugh. Mini mental state A practical method for grading the cognitive state of patients for the clinician. *Journal of Psychiatric Research*, 12(3):189–198, 1975.
- [Fra16] K. C. Fraser and F. Meltzer, J. A. Rudzicz. Linguistic features identify Alzheimer’s disease in narrative speech. *Journal of Alzheimer’s Disease*, 49(2):407–422, 2016.
- [Goo83] H. Goodglass and E. Kaplan. Boston diagnostic aphasia examination. Lea & Febiger, 1983.
- [Goo16] I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning*. MIT Press, 2016.
- [Hoc97] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997.
- [Hus06] A. Husband and A. Worsley. Different types of dementia. *The Pharmaceutical Journal*, (277):579–582, November 2006.

- [Kem89] S. Kemper, D. Kynette, S. Rash, K. O'Brien, and R. Sprott. Life-span changes to adults' language: Effects of memory and genre. *Applied Psycholinguistics*, 10(1):49–66, 1989.
- [Pri16] M. Prince, A. Comas-Herrera, M. Knapp, M. Guerchet, and M. Karagiannidou. *World Alzheimer Report 2016*. Alzheimer's Disease International (ADI), London, 2016.
- [Rab08] L. Rabiner and B.-H. Juang. Historical perspective of the field of asr/nlu. In *Springer Handbook of Speech Processing*, pages 521–538. Springer Berlin, 2008.
- [Rah18] H. Rahmati. *Automatic Classification and Analysis of Task-Oriented Alzheimer Speech*. Master's thesis, Friedrich-Alexander Universität Erlangen-Nürnberg, Erlangen, Januar 2018.
- [Rap11] A. M. Rapp and B. Wild. Nonliteral language in Alzheimer dementia: A review. *Journal of the International Neuropsychological Society*, 17(2):207–218, 2011.
- [SO12] A. L. Sosa-Ortiz, I. Acosta-Castillo, and M. J. Prince. Epidemiology of dementias and Alzheimer's disease. *Archives of Medical Research*, 43(8):600–608, November 2012.
- [Sto02] A. Stolcke. Srilm-an extensible language modeling toolkit. In *Seventh international conference on spoken language processing*, pages 901–904, Denver, 2002.
- [Sto11] A. Stolcke, J. Zheng, W. Wang, and V. Abrash. Srilm at sixteen: Update and outlook. In *Proceedings of IEEE automatic speech recognition and understanding workshop*, volume 5, 2011.
- [Sun12] M. Sundermeyer, R. Schlüter, and H. Ney. Lstm neural networks for language modeling. In *Proc. Interspeech*, pages 194–197, Portland, OR, USA, 2012.
- [Sun14] M. Sundermeyer, R. Schlüter, and H. Ney. rwthlm – the rwth aachen university neural network language modeling toolkit. In *Proc. Interspeech*, pages 2093–2097, Singapore, 2014.
- [Wan17] S. Wankerl, E. Nöth, and S. Evert. An n-gram based approach to the automatic diagnosis of Alzheimer's disease from spoken language. In *Proc. Interspeech*, pages 1572–1576, Stockholm, 2017.
- [Wei08] M. F. Weiner, K. E. Neubecker, M. E. Bret, and L. S. Hyman. Language in Alzheimer's disease. *The Journal of clinical psychiatry*, 69(8):1223–1227, 2008.

- [Zho02] Z.-Y. Zhou, J.-F. Gao, and E. Chang. Improving language modeling by combining heterogeneous corpora. In *International Symposium on Chinese Spoken Language Processing*, Taipei, Taiwan, 2002.